World Development 133 (2020) 105002

Contents lists available at ScienceDirect

World Development

journal homepage: www.elsevier.com/locate/worlddev

Why is household electricity uptake low in Sub-Saharan Africa?

Moussa P. Blimpo [a], Agnieszka Postepska [b,*], Yanbin Xu [c]

[a] World Bank, United States
[b] University of Groningen, The Netherlands
[c] Georgetown University, United States

ARTICLE INFO

Article history:
Accepted 20 April 2020
Available online 21 May 2020

JEL classification:
O13
018
O19
H540

Keywords:
Electricity access
Uptake
Sub-Saharan Africa
Demand side barriers

ABSTRACT

This paper uses recent individual and household-level data from various sources covering over 30 countries in Sub-Saharan Africa, along with qualitative fieldwork in selected countries, to shed light on demand-side constraints to electricity access. First, it uses a simple accounting identity to decompose the electricity access gap into supply, demand, and mixed demand–supply factors. Results show that the demand factors account for over two-fifth of electricity access deficit across the region whereas electrification efforts have largely focused on supply issues. Second, the paper combines regression and qualitative fieldwork analysis to examine household and community characteristics that affect electricity uptake. It finds that at the household level, while the level of income is a primary driver of uptake, the regularity of income is as important. The quality of housing is also a consistently significant predictor. At the community-level, the results show that higher uptake is associated with communities that are better endowed with other infrastructures including public services. This finding is consistent with qualitative evidence supporting that households associate electricity access primarily with the prospect of more productive economic activities. Policies focusing on lowering access costs, streamlining flexible bill payments and connection mechanisms, and efforts to target, or bundle with, complementary income-generating inputs should go hand in hand with supply-side efforts.

© 2020 Elsevier Ltd. All rights reserved.

1. Introduction

Electricity is a pillar of modern economies since it is necessary for most economic activities. While the vast majority of countries outside of Sub-Saharan Africa (SSA) have reached universal access to electricity, only 43% of the population in SSA had access to grid electricity in 2016[1] (IEA, 2017). The electrification agenda is a top priority for governments and populations all across Africa. Likewise, the international community committed through the United Nations' Sustainable Development Goal # 7 (SDG7) to *Ensure access to affordable, reliable, sustainable and modern energy for all by 2030*. The efforts toward SDG7 have largely focused on infrastructure financing and a range of related supply-side issues.[2]

Much less attention has been given to demand considerations, which, if not accounted for, may threaten the financial sustainability of the sector. For example, In Tanzania, a recent massive rural electrification program resulted in an increase of electrified rural villages from 2.5% in 2010 to 7% in 2013. However, only about 3.8% of rural households were connected to electricity in 2013 and the uptake in newly electrified areas was far below the projected numbers (Chaplin et al., 2017). Similarly, Lee et al. (2016) found limited uptake in western Kenya both by households and businesses up to five years after the electric grid reached the communities. While these examples underscore the importance of the

* Corresponding author at: University of Groningen, Department of Economics and Business; Economics, Econometrics & Finance, Nettelbosje 2, 9747 AE Groningen, The Netherlands.
E-mail addresses: mblimpo@worldbank.org (M.P. Blimpo), a.postepska@rug.nl (A. Postepska), yx76@georgetown.edu (Y. Xu).

[1] A household is defined to have access to grid electricity if it is connected to the grid. Access rate is defined as the share of households connected to the grid out of all households.

[2] The supply-side issues are related to the energy mix and climate change, the scale and efficiency of power systems, the inadequacy of investments in the sector (Rosnes & Shkaratan, 2011; Eberhard, Rosnes, Shkaratan, & Vennemo, 2011), poor energy sector performance (Trimble et al., 2016), poor institutional, regulatory, and technical capacity of African power sectors (Ahlborga, Boräng, Jagers, & Söderholmd, 2015; Haanyika, 2006; Karekezi & Kimani, 2002; Blimpo et al., 2017), and power trade (Khennas, 2012).

https://doi.org/10.1016/j.worlddev.2020.105002
0305-750X/© 2020 Elsevier Ltd. All rights reserved.

demand side constraints, little is known about the scale and the drivers of these issues across the continent. To fill this gap, this paper addresses the following two main questions: (i) what is the significance of demand considerations in household electrification across SSA? And (ii) what are the drivers and policy levers to increase uptake on the continent?.

To address these questions we use a mixed-method approach by allowing quantitative analysis and qualitative fieldwork to mutually inform each other. First, we use the World Bank's Living Standard Measurement Survey data (LSMS) to draw basic correlates of electricity access rates among communities to help formulate the basis of qualitative fieldwork in twelve countries where we spoke to households, businesses, and community leaders. We then built on the findings from the fieldwork to sharpen our priors and formulate the final empirical analysis using the Afrobarometer data, the Demographic and Health Survey data (DHS), and the LSMS data.

The qualitative work was conducted in two manners. For the first group of countries (Ethiopia, The Gambia, Nigeria and Senegal) we used focus groups and open interviews, ranging from small to large groups of members of communities. Some of the interviews targeted exclusively the community leaders and chiefs. The sampling followed a version of purposive sampling methodology whereby we continued the interviews and focus groups up to when the conclusions became repetitive. In the remaining countries (Cameroon, Gabon, Ghana, Ivory Coast, Niger, Nigeria,[3] Rwanda, Togo, and Zimbabwe) we sampled five to ten households and we conducted semi-structured interviews, covering the connection cost and process, as well as an open discussions as in the first group of countries.

The quantitative analysis starts with a method developed in Wodon, Banerjee, Diallo, and Foster (2009) to decompose the access deficit into supply, demand, or mixed supply–demand factors. The method accounts for the fact that a lack of electricity infrastructure in a community (off-grid communities) does not mean households in that community lack access only because of supply. That is unlikely to be correct because access rates are not 100% where the infrastructure exists. Similarly, the existence of infrastructure in a community does not imply that all unconnected households lack service due to demand issues. It may also be because many households in covered areas are still located too far from the nearest electric pole, rendering their lack of access partly a supply issue or a mixed demand–supply issue.

Following the decomposition and having shown the importance of demand constraints, we examine the determinants of uptake, focusing primarily on the leading variables that emerged from the qualitative analysis as the main independent variables of interest. If one restricts the data to serviced households and conducts a simple regression analysis controlling for key contributing factors to uptake, the findings cannot be meaningful beyond those areas. A version of that regression is a required first step in the decomposition analysis described above. To be able to infer the findings on areas currently off-grid as well, we adopt Heckman (1976) selection model to account for the fact that the electricity infrastructure roll out is not exogenous to communities' characteristics.

In terms of findings, taking all the analysis together, we report three main findings.

(i) *Households associate electrification primarily with economic activities.* A striking commonality arising from the qualitative research is that households think primarily about income-generating activities when asked about the reason why they would need electricity in their household or in their community. We encountered communities whereby farmers have committed the efforts necessary to power their farms in order to pump water for off-season farming whereas their houses lack electricity. These findings are at odds with the international discourse where electricity for the poor is often thought of in terms of lighting or direct improvement of the living conditions. When it comes to the main constraints and priorities for households, the key variables that emerged concern housing quality and the irregularity or unpredictability of their earnings.

(ii) *The decomposition exercise showed that demand constraints are as important as supply issues.* Across 31 countries in SSA, we find that demand-side constraints can explain 41% of the access deficit. Specifically, demand-side problems are more conspicuous in urban areas where they account for 78% of the deficit. However, they cannot be neglected in rural areas where 35% of the access deficit is explained by demand constraints, despite the lack of electric infrastructure.

(iii) *Household income levels and flows, housing quality, and the reliability of electricity are prime drivers of uptake.* In terms of barriers to access, the regression analysis supports the findings from the qualitative work, namely that the affordability (income), the income flow,[4] the housing quality,[5] and the importance of income-generating activities through the presence of complementary infrastructure in the community are key levers to address the uptake gap. The analysis identified other factors that did not come up as often from the qualitative work. The most prominent of all is the reliability and the quality of the electricity access.[6] The latter finding speaks to a growing literature on welfare of households and unreliable access to electricity in Africa (Lenz, Munyehirwe, Peters, & Sievert, 2017; Dasso, Fernandez, & Nopo, 2015; Lipscomb, Mobarak, & Barham, 2013).

With these findings, the paper contributes to the literature by expanding the body of knowledge on demand for electricity in SSA. It analyzes, at a larger scale, the important issue of uptake. To the best of our knowledge, Lee et al. (2016) and Chaplin et al. (2017) are the two notable exceptions that analyze the same issue, but in a narrower context as described earlier. Both of these studies raise these important questions on demand side constraints.[7] We

[3] In Nigeria, both types of qualitative work were conducted.

[4] The World Bank's Africa Infrastructure Country Diagnostic Study (Banerjee et al., 2008) utilized a pool of household surveys spanning 32 SSA countries and a period of 15 years to analyze the affordability of monthly bills and found that around 60% of the African population cannot afford to pay the equivalent of generation cost (full cost recovery tariffs) or extend consumption beyond the subsidized level (lifeline).

[5] In South Africa, an initiative was put in place to transfer 1.7 million residents of informal settlements to live in formalized housing before electrification could take place (Niez, 2010). The country electrification rate is almost 100%.

[6] The reliability of electricity in SSA is a major concern for policy makers, people, and businesses. Rigorous analysis of this issue is thin but growing. Many papers have assessed the impact of unreliable electricity on the economy, often relying on the frequency of blackout as the main measure of reliability (Blimpo et al., 2018; Mensah, 2018). A recent paper went beyond the blackouts, to provide a close-up look at the quality of voltage and the effect on household and businesses on the Island of Zanzibar in Tanzania (Jacome et al., 2019). However, the link between reliability and uptake has seldom been emphasized as we do in this paper.

[7] Many other studies have examined the uptake of low-capacity solar products (Coville, Orozco-Olvera, & Reichert, 2019; Grimm, Munyehirwe, Peters, & Sievert, 2016; Bensch, Grimm, Huppertz, Langbein, & Peters, 2018). However, many of the solar products marketed do not meet the capacity comparable to grid electricity. To draw a contrast with that literature will require the definition of *electricity* to vary for different households. Studies that have compared grid and solar solutions tend to find that these two are not substitutes (Abdullah & Jeanty, 2011; Lee, Miguel, & Wolfram, 2016). We therefore see it as a separate strand of literature. In a recent note, Peters, Sievert, and Toman (2019) assessed the feasibility of micro-grids that can provide a higher capacity for productive use and identified over-optimistic demand projections as constraint, underscoring further the necessity to address these constraints with regards to electricity for development.

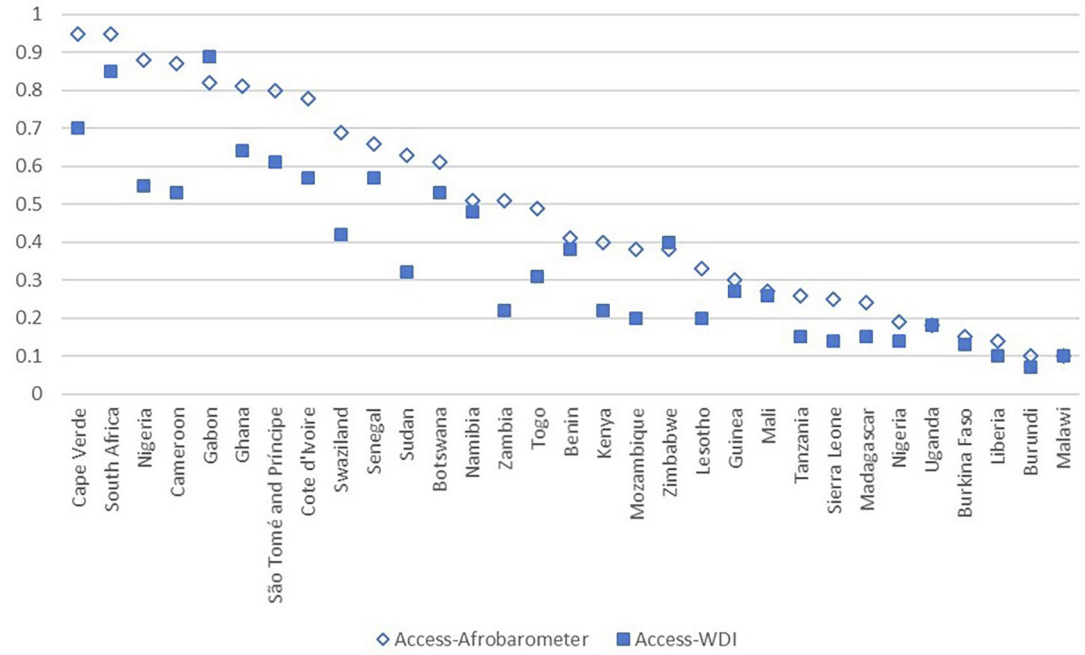


Fig. 1. Electrification in Sub-Saharan Africa. (*Data Source*: Afrobarometer, 2014/15, WDI,2015.).

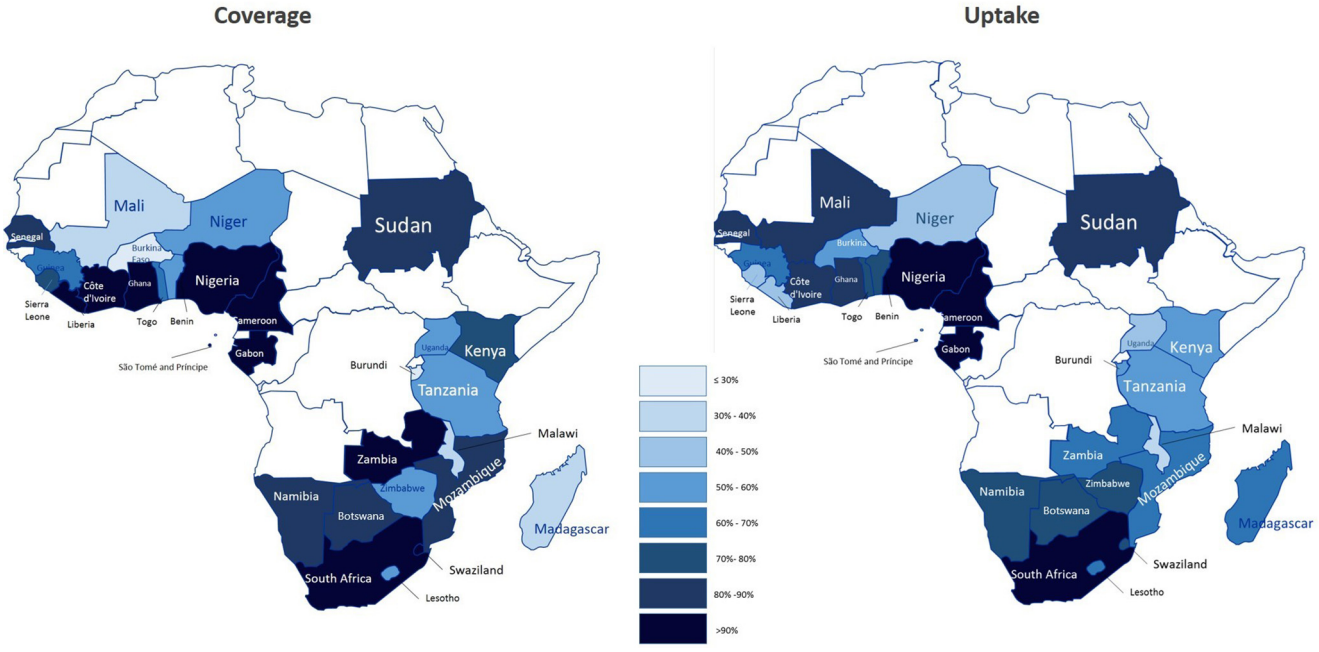


Fig. 2. Coverage and Uptake in Sub-Saharan Africa. (*Data source*: Afrobarometer, 2014/15.).

add to this literature by documenting the importance of demand side constraints in over thirty countries in SSA.

In terms of contribution to policy dialogue, our results add subtle factors beyond income levels, such as household income unpredictability, housing quality, along with households' desire and ability to engage in income-generating activities as likely drivers of uptake. While electrification projects conducted by both national and international organizations, often target access numbers, uptake and demand considerations have not systematically been analyzed and incorporated into the design ex ante. Thus, these results contribute to better policy formulation with respect to the electricity infrastructure planning and roll out.

Hereafter the paper proceeds as follows. Section 2 describes the data used in our analysis. Section 3 provides the context. Section 4 discusses the research methodology. Section 5 presents and dis-

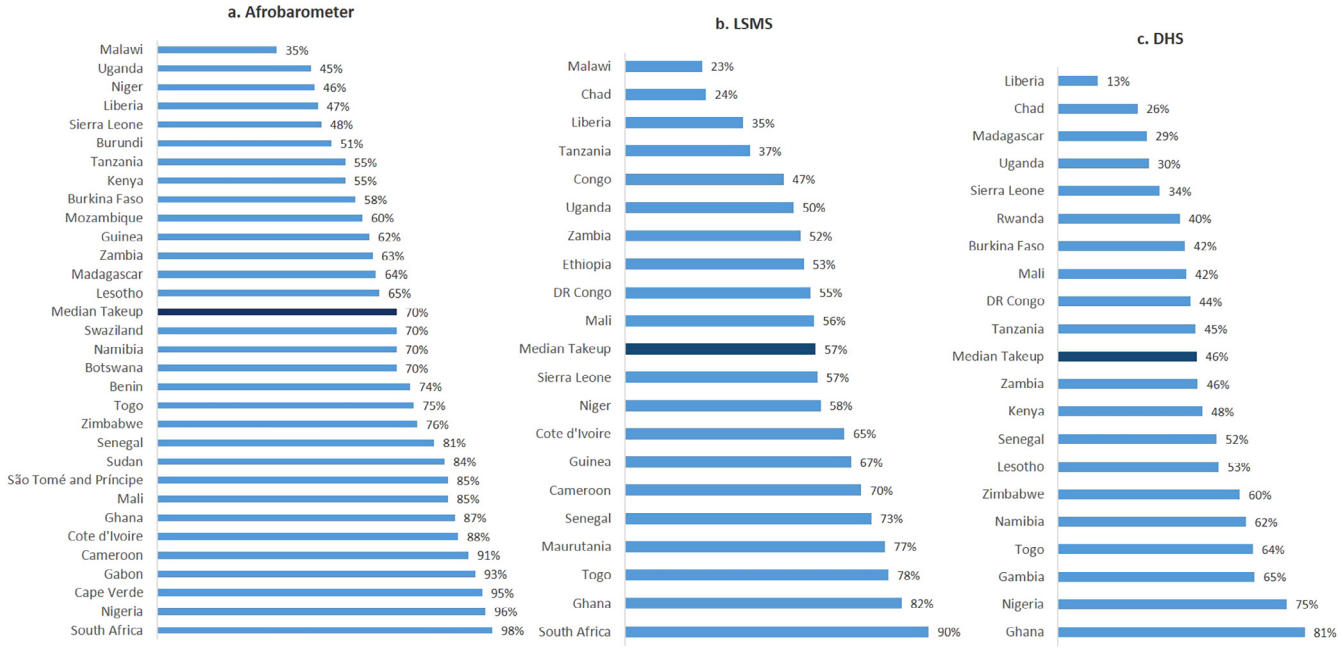


Fig. 3. Electricity uptake using different data source. (*Data source*: Adapted from Afrobarometer, LSMS, and DHS.).

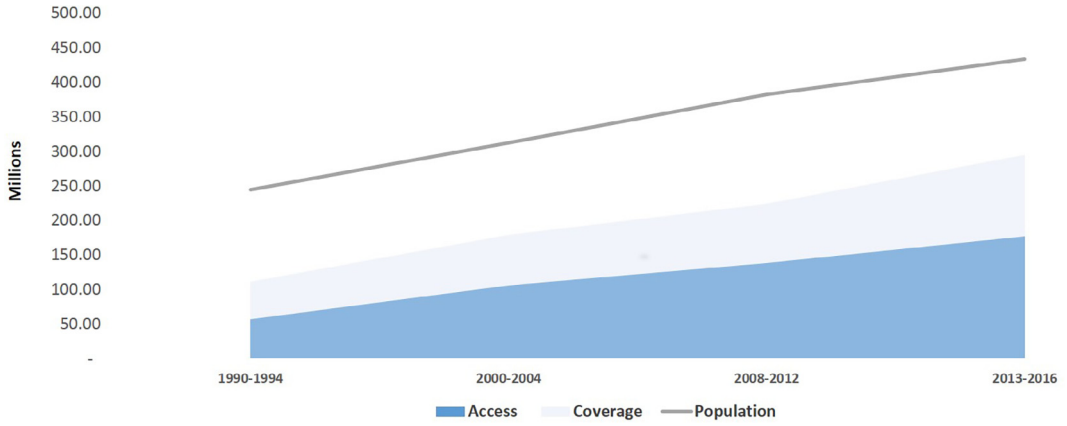


Fig. 4. Coverage, Access and Population Growth Over time. (*Data source*: Adapted from DHS, 1990–2016. Note: The figure shows the aggregated total population, population living under the grid (coverage), and population connected to the grid (access) in Burkina Faso, Ghana, Kenya, Madagascar, Mali, Nigeria, Rwanda, Tanzania, Uganda, and Zimbabwe. The population of the 10 countries combined accounts for, on average,45% of the total population in Sub-Saharan African countries over the years.).

cusses the results. Section 6 concludes and discusses some policy implications.

2. Data

Our analysis relies primarily on the data from Afrobarometer with supplementary data sources from LSMS Panel Data, DHS, Mo Ibrahim Index on Africa Governance and World Development Indicators.[8] To better understand the causes of the demand deficit, we supplement the analysis with data from qualitative study (field work).

2.1. The Afrobarometer data

The primary data set employed in this study is the Afrobaromenter, which measures the social political and economic matters in Africa by conducting face-to-face interviews with a randomly selected sample of 1200 or 2400 people in each surveyed country. The survey is conducted in more than 30 African countries and repeated on a regular cycle. Data from round 6 (2014/2015) is used for the analysis, covering a total of 6055 communities across 31 SSA countries.

As a robustness check, we compare access rates obtained from Afrobarometer to the access rates obtained from 2012 World Development Indicators (WDI) (Fig. 1). The correlation between the two series is 0.832. This is quite high given the differences between the two measurements,[9] including the fact that in the Afrobarometer we define access as the percentage of households connected to the grid whereas in WDI it is the percentage of individuals that have electricity in their homes regardless of the source (WBG, 2012). As part of our analysis is conducted at a community

[8] A detailed description of these data sources is presented in the appendix.

[9] The differences between the Afrobarometer survey data and the WDI data include the years of the survey (2014 for Afrobarometer and 2012 and earlier for the WDI) and the unit of measurement (a household in Afrobarometer and an individual in the WDI).

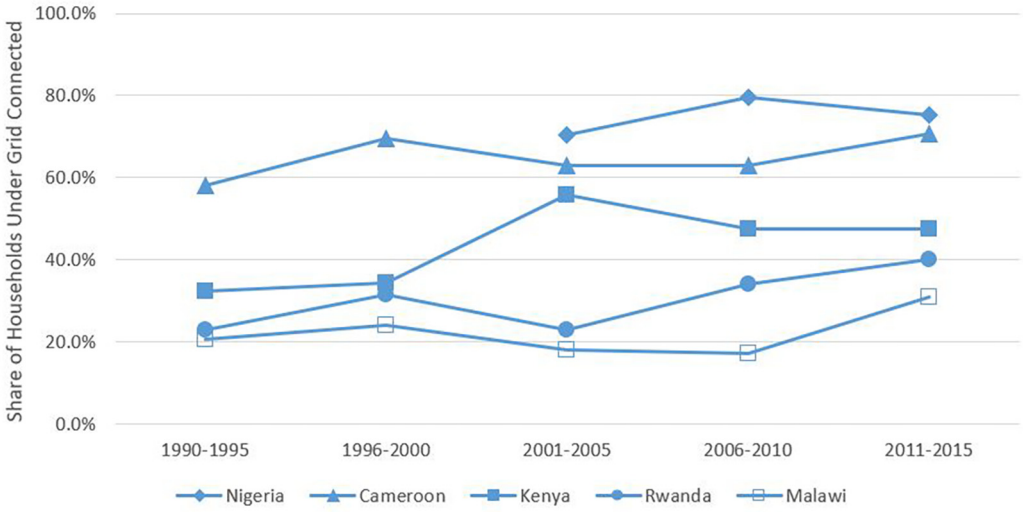


Fig. 5. Electricity uptake over time in selected countries. (*Data Source*: Adapted from DHS,1990–2016.).

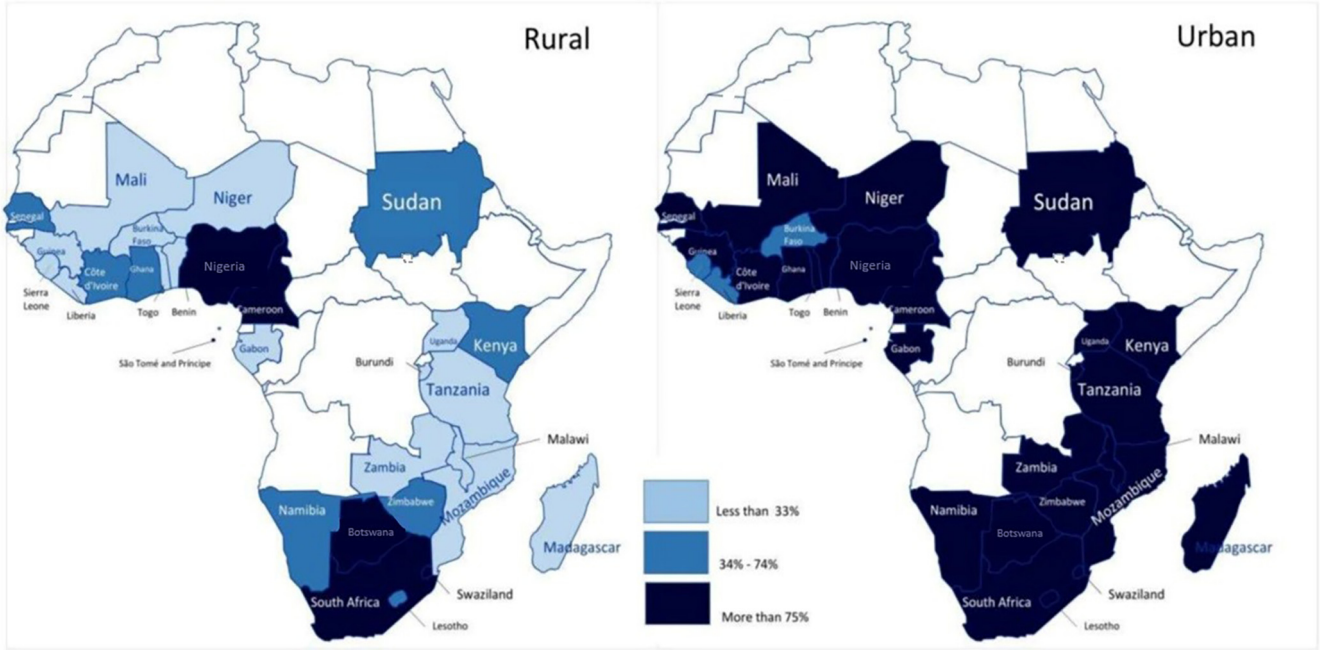


Fig. 6. Grid coverage in urban and rural areas. (*Data source*: Afrobarometer, 2014/15.).

level, we define a community at the primary sampling units (PSU) level. Even though policy makers do not target PSU's but well defined administrative units, using a higher level of aggregation would introduce noise in the measure of access at a community level for each household.[10] If at least one household in the PSU is connected to the electricity grid, we assume this community is covered by the grid. Likewise, if no household in the PSU reports connection to the grid, we label this PSU as an uncovered community. We deem that a household has access to electricity when such household is connected to grid. Uptake refers to households that live in the covered community and connect to the grid. In our sample, 3782 communities are currently covered by grid.

2.2. Qualitative data

To deepen our understanding of why households demand electricity and to better understand the results from quantitative analysis, qualitative field work was conducted in twelve SSA countries: Cameroon, Ethiopia, Gabon, The Gambia, Ghana, Ivory Coast, Niger, Nigeria, Rwanda, Senegal, Togo and Zimbabwe. The countries were chosen to cover different sub-regions of the continent, taking into account the logistical constraints for the feasibility of the survey.

[10] Four PSU's span over more than one administrative level and they are dropped from our sample.

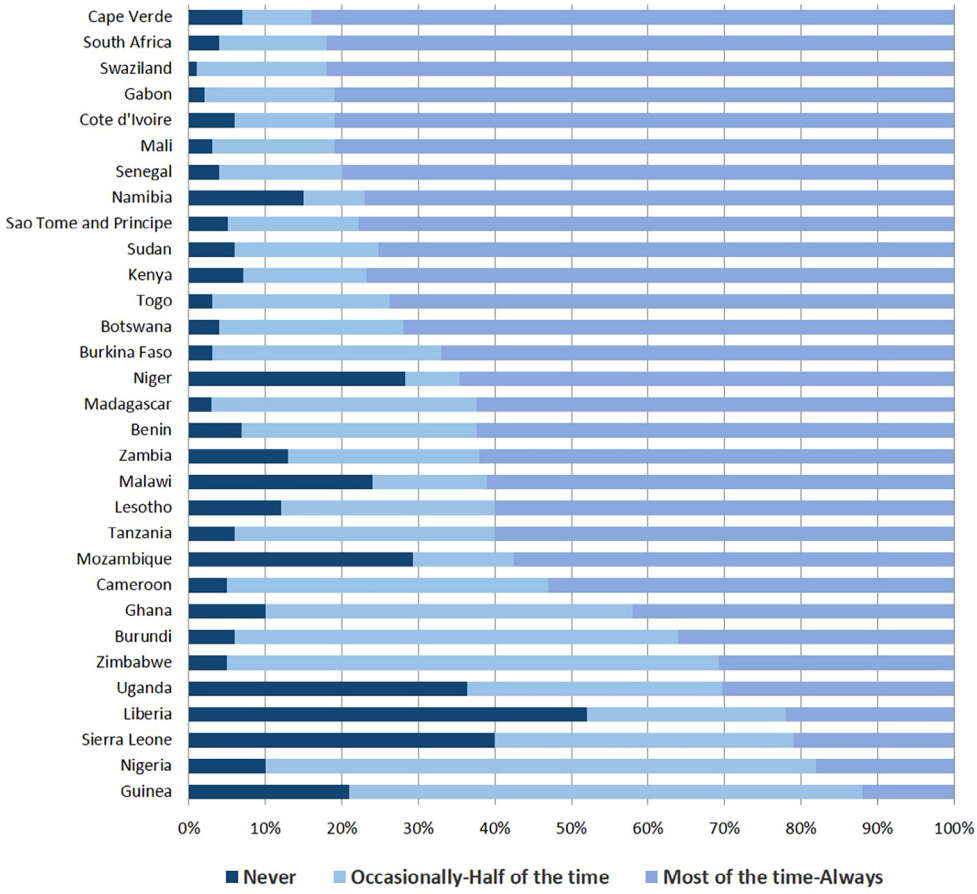


Fig. 7. Reliability of grid electricity in connected households in Sub-Saharan Africa. (*Data source*: Afrobarometer, 2014/15.).

For data accuracy, we focused on recently connected household. We adopted a Theoretical sampling framework, which requires building interpretative theories from the emerging data from early interviews (Marshall, 1996; Kielmann, Cataldo, & Seeley, 2012). Trained enumerators conducted the focus groups that comprised of representatives from utility companies, chiefs of villages, community organizers, residents of the community, university students, and in one case a group of enumerators (in Senegal) who conducted energy survey in dozens of communities. Topics covered during the interviews often started with the need for electricity, the affordability and then branched out into appliances acquisition, productive use, reliability, and many more subjects.

3. Contextual background

3.1. Low uptake

Fig. 2 illustrates coverage[11] and uptake[12] for selected SSA countries and reveals a great variation along the two measures across the countries. Coverage rates vary from 21% in Burundi to 99% in Cape Verde indicating very different contextual landscapes across countries. Uptake rates also vary greatly. While in Malawi only 35% of households under the grid are connected, in South Africa almost every household enjoys grid connection in the covered areas.

Majority of SSA countries are characterized by coverage below 80% indicating a substantial need for infrastructure provision. In Burundi one in every five households is located in a covered area. In Burkina Faso, Malawi, Mali, Madagascar, Tanzania, Lesotho, and Zimbabwe less than or barely 50% of households are covered by the grid indicating that less than one in every two households has connection to the gird.

While the disparities in coverage among countries in terms of infrastructure development are expected, the variations among uptake appear to be less straightforward. Due to different methodology of measurement, uptake differs among data sources (depicted by Fig. 3). However, it bolsters the point that few countries enjoy high uptake rates, which underscores the significance of demand-side barriers to electricity access. Cape Verde is the only country with both very high coverage and uptake rates, with over 90% of all households having grid connection. In many other countries, even though the coverage is high, the uptake rates remain relatively low implying that the lack of connection is not solely due to supply-side issues. The low uptake problem is unlikely to be resolved by the supply-side solutions. Fig. 4 shows that while the grid coverage in SSA has been extended over the past 30 years (particularly after 2010), it could not outpace the growth of population. Therefore, population with access has grown very slowly, which is reflected by the flat trend line in Fig. 4. Fig. 5 depicts uptake figures over time in Nigeria, Cameroon, Malawi, Rwanda, and Kenya. The figures are higher in Nigeria and Cameroon but still remain below 80%. Rwanda and Malawi have shown significant gains recently from a very low base. In Kenya uptake increased in the decade before 2005 and declined afterwards. The decline is indicative of the demand-side issues becoming a more

[11] Coverage is defined as the share of households located in areas serviced by grid electricity.

[12] Uptake is defined as the share of households in the serviced areas that are connected to the grid.

significant constraint, as the country expanded the grid to more rural and impoverished areas (Lee et al., 2016). It also appears that demand-side constraints are more complex than merely affordability issues. For example, out of countries where majority (no less than 75%) of the areas are covered by the grid, Kenya has the lowest uptake rate (42%). If it is all due to affordability, then countries with similar per capita income should enjoy equally low uptake rates, controlling for the prices. Kenya's GDP per capita is about 2818 USD (in 2011 USD). Sao Tome and Principe and Cameroon are both characterized by the same level of coverage as Kenya and have similar GDP per capita of 3030 USD and 2835 USD respectively. However, the uptake in both countries is much higher than in Kenya (85% and 91% respectively). This evidence suggests presence of other demand-side barriers to electricity access and indicates that conventional grid expansion efforts might fail at achieving universal access to electricity. Recent study on Burkina Faso confirms this conjecture by concluding that "*the current grid extension is becoming inefficient and unsustainable in order to reach the national energy access targets*" (Moner-Girona, Bódis, Huld, Kougias, & Szabó, 2016).

3.2. Large disparities between rural and urban areas

Splitting the sample into rural and urban areas reveals that rural areas are at a great disadvantage in terms of grid coverage as illustrated in Fig. 6. While almost all countries enjoy at least 70% coverage in urban areas, in the rural areas, only 7 countries in our sample enjoy similar coverage. Rural SSA, with the exception of Cape Verde, Swaziland, South Africa, Botswana, Cameroon, Nigeria remains broadly without grid coverage.

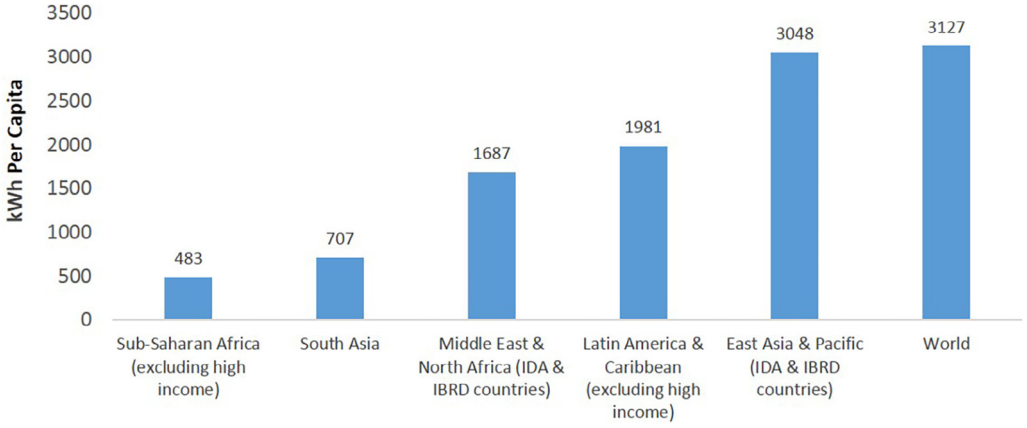


Fig. 8. Electric power consumption, 2014 (kWh per capita). (*Data source*: WDI, 2014.).

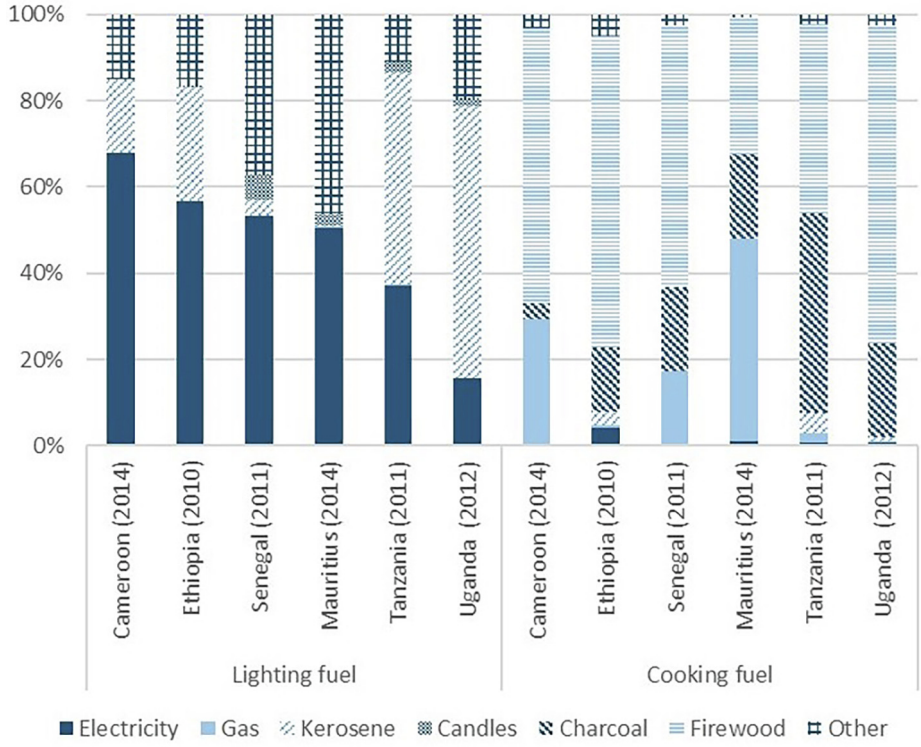


Fig. 9. Cooking and lighting fuel in selected countries. (*Data source*: Adapted from LSMS, 2010–2016.).

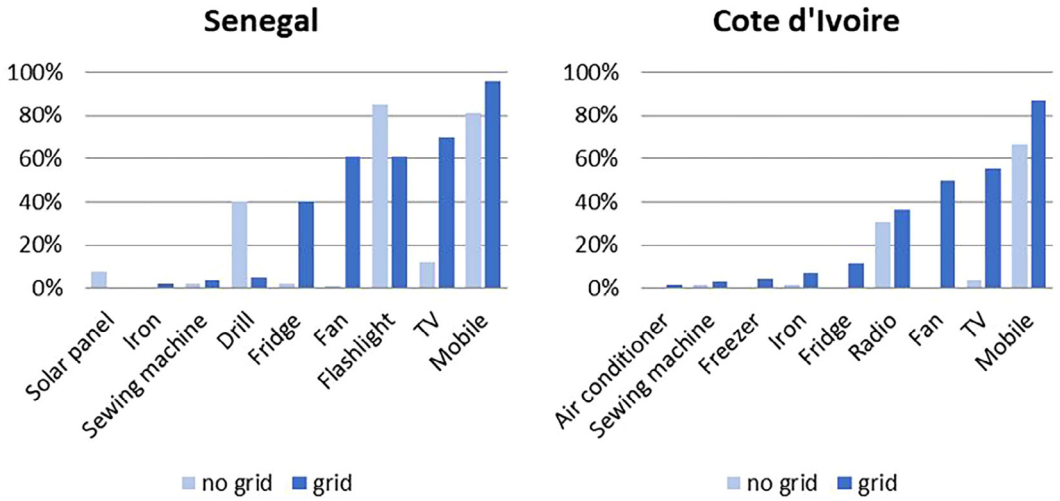


Fig. 10. Appliances possession by grid connection in selected countries. (*Data source*: Adapted from LSMS, 2010–2016.).

3.3. Unreliable electricity services

So far we have classified households based on a simple binary connection indicator. However, the necessity to redefine access to modern electricity is highlighted by recent literature (CGD, 2017). If a household experiences frequent and long lasting blackouts, should we consider it as having access?.

Fig. 7 shows a dramatic variation across countries in terms of reliability of main grid electricity. In Liberia, over 50% of connected households report that they never have electricity. Sierra Leone, Uganda, Mozambique and Niger all appear to have severe reliability problem with over or close to 30% of connected households reporting never having electricity. Cameroon, Ghana, Burundi, Zimbabwe, Nigeria and Guinea have near 50% of connected households with electricity available occasionally. On the other hand, Cape Verde, South Africa, Swaziland, Gabon, Cote d'Ivoire, Mali and Senegal all appear to have relatively reliable supply of electricity with at least 80% of households having access at least most of the time. Nigeria represents a stark example as it enjoys almost 100% coverage rate. This would deceptively leave us thinking that the goal of universal electrification, at least in the urban areas, has been reached in Nigeria. However, less than 20% of households report having electricity most of the time and 51% of households report having electricity occasionally in the dwelling. It remains an open question whether such households should be classified as households with access to modern energy. Another issue not captured by the data at hand, but which became apparent during the field work in Nigeria, is voltage availability. The electricity available to households might be sufficient to light up a low voltage light bulb but is not enough to power even a small appliance such as a fan.

3.4. What do households demand electricity for?

Electric power consumption in SSA is extremely low compared to other developing regions (Fig. 8). The 483-kilowatt hours (kWh) per person consumed in SSA in 2014 is not much more than the amount of electricity needed to power a 50-watt light bulb continuously for a year (IEA, 2014). If SSA countries were to connect all households quickly, the average level of consumption would remain low, because most cannot afford electrical appliances such as air conditioners, refrigerators, and electronic heaters. Given the prevalence of subsidies, the financial situation of the utilities operating in the region would worsen, threatening their sustainability. It is therefore imperative that as the region makes progress toward universal electricity access, consumption of electricity also rises. Given the empirical evidence, it is not surprising to see that households utilize various sources of energy. Energy stacking is a common strategy among African households in which they do not rely solely on one energy source but on a portfolio of sources (van der Kroon, Brouwer, & van Beukering, 2013; Kowsari & Zerriffi, 2011). Fig. 9 summarizes the usage of different cooking and lighting fuels in selected countries. For example, while in Cameroon, 68% of households use electricity as the main lighting fuel, almost no household uses electricity for cooking (around

Table 1
Households determinants of uptake: Probit model results.

	All	Urban	Rural
% connected in PSU	3.318***	3.592***	3.153***
	(72.356)	(44.760)	(54.267)
Urban	0.081**		.
	(3.249)		.
Household size	0.040***	0.045***	0.041***
	(7.960)	(5.100)	(6.671)
Wealth quintiles			
2	0.323***	0.382***	0.064
	(12.028)	(8.369)	(1.582)
3	0.490***	0.647***	0.243***
	(16.287)	(11.465)	(6.047)
4	0.720***	0.743***	0.454***
	(19.461)	(10.474)	(10.193)
5	0.988***	1.012***	0.771***
	(23.652)	(11.590)	(16.320)
High quality roof	0.130***	0.319***	0.120***
	(4.448)	(4.876)	(3.610)
Secondary education	0.264***	0.382***	0.203***
	(9.942)	(9.021)	(5.790)
Cash paying job	0.077**	0.046	0.118***
	(2.905)	(1.091)	(3.458)
% in agriculture in PSU	−0.107***	−0.143*	−0.126***
	(−3.771)	(−2.409)	(−3.849)
% professionals in PSU	0.082*	0.109*	0.083
	(2.524)	(2.217)	(1.884)
Age	0.000	−0.002	0.001
	(0.150)	(−1.485)	(1.457)
Female	0.033	0.016	0.032
	(1.581)	(0.470)	(1.189)
Constant	−2.399***	−2.534***	−2.282***
	(−28.132)	(-20.055)	(-17.739)
N	33479	17600	15879

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
Notes: Coefficients are reported; *t* statistics in parentheses. The table reports the estimates of Eq. 2 for the "All", "Urban" and "Rural sample". *Data source*: Afrobarometer.

Table 2
Demand supply decomposition for Sub-Saharan Africa. Comparison of different constraints.

	PDSG	SSG	PSSG	MDSG	% deficit "demand"	% deficit rest
Demand Gap as a Whole (statistical approach)						
All Africa	0.19	0.27	0.20	0.07	0.41	0.59
Urban Africa	0.13	0.04	0.03	0.00	0.78	0.22
Rural	0.23	0.43	0.26	0.17	0.35	0.65
	APDSG				% deficit "demand"	% deficit rest
Wealth quintile						
All Africa	0.09				0.19	0.81
Urban Africa	0.11				0.25	0.75
Rural	0.09				0.21	0.79
Cash paying job						
All Africa	0.07				0.29	0.71
Urban Africa	0.09				0.38	0.62
Rural	0.05				0.27	0.73
Roof quality						
All Africa	0.07				0.29	0.81
Urban Africa	0.09				0.38	0.62
Rural	0.04				0.21	0.79

Notes: PDSG denotes the pure demand side gap, SSG denotes the supply side gap, PSSG denotes the pure supply side gap and MDSG denotes the adjusted mixed demand side gap. APDSG denotes the adjusted pure demand side gap. See appendix for details of computation. The last two columns show the percentage of the access deficit due to demand and other constraints. All estimated gaps are statistically significant at 1% level as assessed using a bootstrap procedure using 500 replications.
Data source: Afrobarometer.

64% of households still use firewood as the main cooking fuel). Kerosene is still the main source of lighting in some countries while a majority of households in Africa rely on charcoal and firewood for cooking.

The low uptake of modern cooking fuels among connected households implies that health hazard related to pollution from burning bio mass does not decrease with electrification. Moreover, more recent study shows that the increased availability of LED lamps decreases the usage of kerosene in rural Africa substantially. The cost of such lamps can be scaled almost continuously (as it is reflected in the number of LED diodes) and therefore is perfectly suited for financially constrained households (Peters & Sievert, 2016). The obvious cost advantages as well as the widespread access to such source of lightning, significantly decreases the role of grid electrification in providing light to African households.

Household's possession of appliances is in principle a good indicator of why households demand electricity. However the information available in the data is very scarce. Only for few of the countries, LSMS data contain detailed information on appliance possession. Consider the appliances households report by electricity connection in Senegal presented in Fig. 10. Not surprisingly households with grid connection are much more likely to have a fridge, a fan, a TV or a modern stove.[13]

Overall, the information available in the quantitative data on usage of electricity is very limited. Even if we know whether a household is in possession of certain appliance, we cannot say anything about its usage. Therefore, we revert to the vast evidence from the qualitative work in twelve SSA countries to better understand the ways in which households utilize electricity. Contrary to the common perception that electricity is primarily demanded for lighting of the dwelling,[14] our evidence from field work suggests that individuals primarily recognize its productive use. The emphasis is often placed on the desire to bring electricity to a place of work before bringing it to the household. Both in Senegal and Nigeria, households utilize electricity for productive activities such as purchasing a fridge to sell cold drinks or purchasing a small electric tool used in welding or carpentry which creates employment opportunities. Even cell phone usage is directly linked to productive activities as it provides main means of communication. In addition, in rural Senegal, individuals emphasize the importance of electricity in agriculture for irrigation purposes. Evidence from India further supports the productive use of electricity (Van de Walle, Ravallion, Mendiratta, & Koolwal, 2013). Examining the long run effects of the Indian roll-out program, Van de Walle et al. (2013) find substantial increase in labor supply both among men and women. The authors find that men shift leisure time to evening hours and work more regular hours during daytime. Among other non-financial factors, the desire of an outside light as a status indicator as well as its importance for safety was often mentioned during the discussions. Increased pubic safety through outside lighting is a well recognized issue in the literature (Kalra, Shekhar, & Shrivastava, 2007).

4. Empirical strategy

4.1. Demand supply decomposition

In order to quantify the deficit in access rate due to the lack of demand, we employ the methodology developed in the study of coverage of modern infrastructure services in African cities by Wodon et al. (2009).[15] The authors extend a framework used to analyze the infrastructure services in Guatemala (Foster & Araujo, 2004) to account for the imperfect measurements of electricity access due to the fact that relying on a purely statistical method as in Foster and Araujo (2004) may lead to a substantial bias. Some households may be located in covered areas but they may be too far from the actual grid. Such household would be mistakenly classified as "not connected" due to demand factors, whereas it is primarily not connected due to lack of supply. Wodon et al. (2009) suggest to use regression to mitigate this type of bias. In our analysis, we diverge from Wodon et al. (2009) and decompose the demand constraints into different components, therefore we cannot talk about demand–supply decomposition as straightforward as in Wodon et al. (2009). Below we briefly describe the methodology noting that we focus on the deficit due to demand constraints.[16]

Let A (Access) denote the share of households using electricity (connected to the grid) divided by the total number of households.

[13] The definition of modern stove combines electric and gas stoves.
[14] In Senegal individuals report that the cost of kerosene is incomparably lower than the cost of lighting powered by electricity.
[15] Note an important change in notation: we define *coverage* as the percentage of households located in serviced areas (out of all households). In Wodon et al. (2009) it is referred to as access. We define *access rate* as the share of serviced households that are connected to the grid.
[16] For detailed discussion see the appendix and Wodon et al. (2009).

Table 3
Deficits for Sub-Saharan Africa. Comparison of different constraints by country sub groups.

Country split	*Scenario 1*			*Scenario 2*			*Scenario 3*		
	Demand	Supply	Mix	Demand	Supply	Mix	Demand	Supply	Mix
LIC	0.32	0.27	0.41	0.30	0.28	0.42	0.28	0.29	0.43
LMIC	0.59	0.22	0.18	0.56	0.24	0.20	0.53	0.25	0.21
UMIC	0.66	0.20	0.14	0.63	0.22	0.15	0.59	0.24	0.17
Africa	0.39	0.30	0.31	0.37	0.31	0.32	0.35	0.32	0.33
Rural	0.34	0.26	0.40	0.32	0.27	0.41	0.30	0.28	0.42
Urban	0.74	0.17	0.09	0.70	0.19	0.11	0.66	0.21	0.12
West Africa	0.38	0.31	0.31	0.36	0.32	0.32	0.34	0.33	0.33
East Africa	0.34	0.25	0.41	0.32	0.26	0.42	0.30	0.26	0.43
Central Africa	0.85	0.12	0.03	0.80	0.15	0.05	0.76	0.18	0.06
Southern Africa	0.55	0.24	0.21	0.52	0.26	0.22	0.49	0.27	0.24
Share of Urban population above median	0.63	0.21	0.16	0.60	0.23	0.18	0.56	0.24	0.19
Share of Urban population below median	0.32	0.28	0.40	0.30	0.29	0.41	0.29	0.29	0.42
Share of Renewable energy above median	0.33	0.32	0.35	0.31	0.33	0.36	0.30	0.34	0.37
Share of Renewable energy below median	0.42	0.28	0.30	0.39	0.29	0.32	0.37	0.30	0.33

Notes: All three scenarios assume that if the grid was to be expanded today to the areas without grid, the uptake would be two-thirds of the uptake in the areas already covered by the grid. In the areas covered by the grid, scenario 1 assumes that still 5% of the access deficit can be attributed to supply-side constraints. Similarly, Scenarios 2 and 3 assume that the supply constraints would account for respectively 10% and 15% of the access deficit in the areas covered by the grid.Data source:Afrobarometer.

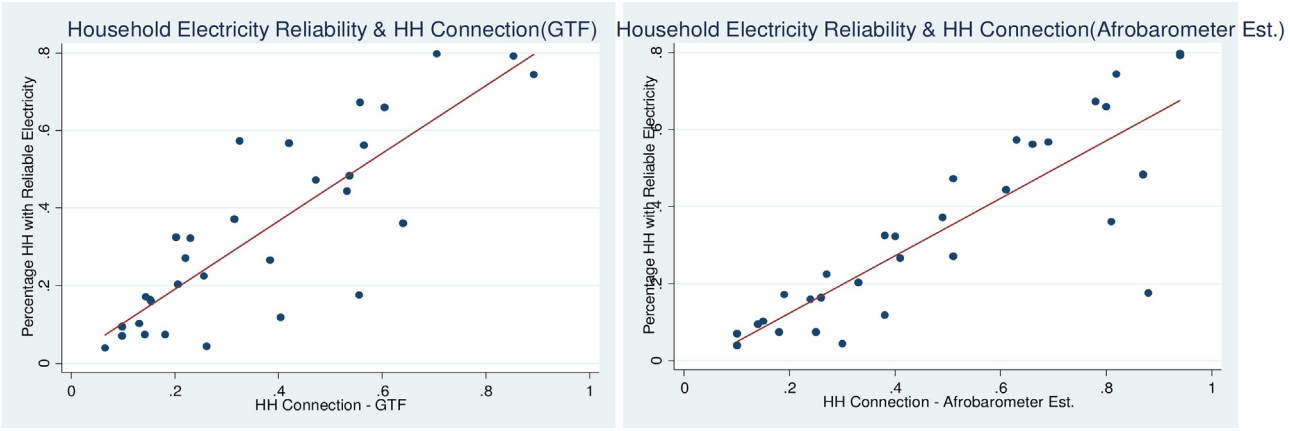


Fig. 11. Household connection rates and reliability of electricity service. (*Data source*: Afrobarometer, 2014/15; Global Tracking Framework, 2014.).

Further, let C (Coverage) denote the share of households located in covered areas and U (Uptake) denote the share of households in the covered areas that are using grid electricity. Therefore, $C \times U = A$. Using the statistical approach to the decomposition, the pure demand-side gap (PDSG) can be defined as:

$$PDSG = C \times (1 - U) \tag{1}$$

where $(1 - U)$ is the share of unserved households in covered areas. The proportion of the deficit in access[17] that is attributed to the demand-side factors can be expressed as $\frac{PDSG}{1-A}$. It is the share of the pure demand gap in the unserved population. To mitigate the bias described above and more accurately estimate the share of the demand side constraints in the access deficit, we turn to the econometric approach as proposed in Wodon et al. (2009).

We first estimate the uptake at a household level from the regression of households' determinants of uptake. As we are interested in the individual uptake in covered areas, we restrict our sample to areas with grid connection and draw inference on households in covered areas, acknowledging the limitation to apply the results beyond covered areas due to the selection bias. We do so for the whole region as the country-level samples result in a relatively small number of observations. In order to accurately estimate the demand gap, we first develop a simple model that will allow us to predict the uptake once certain demand side barriers are removed. The underlying model can be summarized as follows:

$$y_i = I(\mathbf{x}_i\prime\beta + \epsilon_i > 0) \quad i = 1, \ldots, N \tag{2}$$

where y_i is a binary variable taking value 1 if household i is connected to the grid and 0 otherwise, $\mathbf{x}_i$ is a vector of control variables and N denotes the number of households in areas with grid coverage. We select the set of explanatory variables to best capture the key aspects of the demand-side barriers given the information available in the data.

First, we believe that the average connection rate at the community level, measured as the share of connected households in the PSU, plays an important role. On one hand it is a good proxy of the connection cost. On the other hand it conveys useful information about potential peer effects. If more households are connected to electricity, then the benefits (or lack of such) from connection are more visible to other households and might contribute to the increase (or decrease) in overall uptake rate. Spil-

[17] In Wodon et al. (2009) it is referred to as deficit in coverage due to differences in definitions.

lover effects in adopting new technologies in developing countries are well documented. The evidence from literature on agriculture in Africa suggests that individuals are more likely to mimic individuals who are similar to them. Another factor at a community level is the economic profile of a locality as it partially captures the productive use of electricity.

Regarding household level characteristics, an important concern is household income. We control for household's wealth measured by the wealth index,[18] which captures household's possessions, amenities and economic situation. It is a well known fact that financial constraints play a major role in determining whether a household connects to the grid. Little however is known about the attributes of the households income and its impact on the decision to connect. We aim to shed light on this issue by focusing on the regularity and predictability of income to capture the household's ability to pay monthly bills. This is measured with the dummy variable indicating whether at least one households member has a full-time or part-time cash paying job. This is especially important for households in the lowest wealth quintile. Moreover, as stated in the introduction, housing quality, measured with a dummy variable indicating that a household has a high quality roof, is a factor that cannot be omitted. Additionally, we control for household size, gender, age and schooling of the respondent and the locality (rural or urban) of the household.[19]

Using the estimates of Eq. 2 we simulate what the uptake would be if the demand side constraints were lifted. We then aggregate the results over all households in serviced areas to obtain the simulated uptake rate U^*. We can then compute the adjusted pure demand side gap (APDSG), which is defined as the difference between the simulated access rate ($C \times U^*$) and the actual access rate ($C \times U$):

$$APDSG = C \times (U^* - U) \tag{3}$$

In Wodon et al. (2009) the demand-side factors are captured by a single variable denoted as the index of household wealth. We utilize the richness of the Afrobarometer data to estimate the model of the determinants of the uptake at a household level and decompose the demand-side factors contributing to low access into different attributes of households to extend the framework proposed in Wodon et al. (2009). We consider three constraints: *affordability*, *cash flow* and *housing quality*. The first constraint is measured with a household wealth index as described and aims to capture differences in economic well being of households. It is already well known that poor households are at a disadvantage when it comes to access to electricity, therefore we do not focus on it in our analysis. The remaining two constraints are of importance as they aim to capture a specific attribute of household income and economic situation.

While the level of income of a household is undoubtedly important, its predictability and regularity (and often its form) are critical when analyzing electricity access. Households are required to sign long-term contracts and often face high re-connection fees and/or penalties for disconnecting. A recurring payment of an even small amount can constitute a major problem for households who generate their income on a non-regular basis. Moreover, while the stock of animals or certain durable goods is an indicator of a household wealth, its lack of liquidity makes it less relevant in our context. *Cash flow* constraint aims to capture this aspect of household wealth. The logic behind the dwelling condition is simple. Two households with the same level of overall wealth potentially face very different needs in terms of housing infrastructure. *Housing quality* tries to measure this aspect of household characteristics. It is measured with roof quality as this information is available for all countries in the sample. Moreover, it appears to be crucial in the context of electricity as some types of shelters by regulation cannot be connected to the grid.

In order to assess the importance of each of the three attributes of demand for the access deficit, we simulate the uptake rate (U^*) conditional on lifting up every constraint of interest in isolation. Therefore, we compute the demand-side gap and the deficit attributed to each specific demand-side constraint separately. The proportion of the deficit in access that is attributed to each of the demand-side constraints can now be expressed as $\frac{(APDSG|\text{demand side constraint})}{1-A}$. It is the share of the adjusted pure demand gap due to the constraint in question in the unserved population.

4.2. Uptake in off-grid areas

To simulate uptake rate in communities not covered by the grid, a new model rather than decomposition should be adopted. First, grid electricity service is absent in these areas, which would give us zero coverage. Second, if communities under and off grid vary significantly in terms of observable characteristics, then they are also very likely to differ along unobserved characteristics. Electricity roll out is certainly not random and failure to account for it leads to inconsistent estimates and therefore, we proceed with the Heckman selection model which can be summarized as follows:

$$\text{Uptake}_c = (\mathbf{x}_c'\theta + \nu_c) \times \text{Access}_c \tag{4}$$

$$\text{Access}_c = I(\mathbf{z}_c'\gamma + u_c > 0) \tag{5}$$

where c denotes a community and $Access_c$ is a dummy variable indicating whether community c is serviced by grid electricity or not. $\mathbf{x}_c$ and $\mathbf{z}_c$ are vectors of community characteristics that can be broadly categorized in three groups. The first group comprises indicators of installed infrastructure such as access to piped water, sewage, cell phone reception and paved roads, as well as presence of a school, a bank, a market, a clinic and a post office. This category also includes the average connection rate as well as the share of individuals with access to reliable electricity which we define as having electricity at least most of the time. The second category contains variables describing socioeconomic status of individuals and households in respective communities, such as the fraction of individuals with secondary education or cash paying job, unemployment rate and the share of households with high quality roof as well as a set of variables capturing the type of economic activities in which members of a community are involved. The latter includes agriculture, trading, unskilled work, skilled work, clerics, supervisors, security, upper professionals and others. The last set of variables composes of macroeconomic indicators at a country level such as GDP per capita, population density, share of population of the same ethnicity as the president, polity score, ethno-linguistic fragmentation, Mo Ibrahim infrastructure and business rural sector scores.

Even though the Heckman selection model is theoretically identified if the set of variables in $\mathbf{x}_c$ and $\mathbf{z}_c$ is the same, to ensure that the model is identified, we consider two exclusion restrictions. Therefore, in the selection equation, vector $\mathbf{z}$ contains all relevant variables that are in $\mathbf{x}$ and additional variables (exclusion restrictions). The obvious suspect for an exclusion restriction is the land gradient (elevation). Land gradient is a critical factor in prioritizing

[18] There is no direct measure of household income in the data. Instead, we construct an index based on the information on household's possessions (TV, radio, motor, vehicle or motorcycle), amenities (presence of toilet and running water), remittances, frequency of lack of cash and subjective assessment of the household's living conditions. Please see appendix A for details

[19] An often overseen issue in related literature is the distinction between individuals in owned and rented dwellings. It is rarely the case that individuals that rent make the decision whether to connect to the grid and therefore it appears to be reasonable to confine the analysis to those living in owned units only. Unfortunately this information is not available in Afrobarometer data.

areas for grid extension since it directly affects the cost of electricity distribution. The less gradient the land has, the cheaper it is to lay power lines, erect transmission poles and eventually reduce the average cost of household connection (Dinkelman, 2011). However, in the literature land elevation is used to instrument individual household connection conditioned on the community having access to the grid. Within a community, a difference in land elevation generates exogenous variation in connection costs. However, elevation at the community level is also correlated with many other factors that determine households' financial viability and thus it is a likely determinant of the connection rate at a community level. Even with a rich set of controls, land elevation remains significant in the main equation and thus we include it as an additional control in the uptake equation.

The first exclusion restriction employed in the model, *contact with government agency*, aims to capture the community connection to the government. It is a dummy variable that takes on value 1 if at least one individual in a community reported contacting a member of a government agency at least once within the last year. As revealed during the field work, significant amount of bureaucracy is often involved in bringing electricity to a community and having a representative in the government wiling to assist a community in obtaining coverage might be an important asset. Therefore, often contact with members of government agencies can directly influence the probability of grid connection at a community level. It is possible however that it also affects other economic activities in the region which can affect the uptake rates. Nevertheless, controlling for such activities as well as for other indicators of an economic livelihood in the main equation reduces this confounding effect.[20]

The second exclusion restriction concerns the main source of electricity. Hydro and gas are the two main sources of electricity in the countries considered. In our analysis, we include an indicator for hydro being the main source of energy. The rationale behind this instrument relies on the fact that the source of energy determines the costs of both generation and distribution. Therefore, it affects the grid expansion but should not have a direct effect on the uptake of electricity among households within the community. As such costs can be reflected in prices, we run a series of regression to see if the main source of generation is reflected in the prices at different levels of electricity consumption and we find that there are no major differences between hydro and gas in terms of electricity prices.[21] We also conduct a robustness check where we include country level tariffs to minimize the potential confounding effect.[22]

In urban areas, 94% of the communities are covered by the grid implying that selection does not present itself as a significant issue. On the contrary, in rural areas less than 50% of communities are currently covered by the grid. While it is certainly important to understand what are the characteristics of the 6% of the urban communities that do not have access to the grid, we believe that qualitative work is better equipped to do so.

[20] There are no standard test that we can use to verify whether the exclusion restrictions are valid. However, we do perform two checks to ensure that it is indeed the case that the exclusion restrictions do not improve the fit of the main equation if included in it directly. We first estimate the model using a two step procedure and we include the exclusion restrictions in the main equation to see if they are significant. Then we estimate the model using maximum likelihood and we perform a likelihood ratio test of our specification against a specification in which the exclusion restrictions are included in the main equation. The results of these tests are presented in the tables with results.

[21] However, note that the sample is small (31 countries).

[22] Other instruments considered for future investigation include distance to generation facility and distance to capital city. However, we cannot include those due to data availability.

Table 4
Comparisons of areas under and off grid.

	Under the grid	Off grid	Difference
% in agriculture	0.06	0.17	-0.11***
% in trading	0.05	0.09	-0.05***
% upper professional	0.09	0.05	0.05***
Pipped water	0.68	0.29	0.39***
Sewage system	0.39	0.01	0.37***
Bank	0.31	0.07	0.24***
Market	0.69	0.55	0.14***
Public transport	0.88	0.64	0.24***
Post office	0.26	0.05	-0.20***
Clinic	0.6	0.46	-0.14***
Paved roads	0.59	0.27	0.32***
% with secondary schooling	0.38	0.1	0.28***
% with cash paying job	0.42	0.27	0.16***
% with high quality roof	0.85	0.53	0.33***
GDP per capita	4956	1721	3236***
Polity index	4.74	4.01	0.72***
% of president ethnicity	0.25	0.40	-0.15***
Ethno linguistic fragmentation	0.72	0.61	0.11***
Mo Ibrahim infrastructure score	41.47	32.96	8.50***
% thinking electricity is imp issue for gov	0.17	0.13	0.03***
% thinking gov is handling electricity well	0.62	0.48	0.14***
% with reliable electricity	0.56	-	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
Notes: Except for the variables denoting coverage and presence of other infrastructures, all other variables are the share of household/residents having that particular attribute. For instance, the secondary schooling at a community level is the share of people who have secondary education in the community. Data source: Afrobarometer.

5. Results and discussion

5.1. Demand supply decomposition

We start with the simple accounting exercise (statistical approach to the decomposition). The results are presented in the top panel of Table 2. The first four columns show the pure demand, supply, pure supply and mixed gap either resulting from the simple statistical approach or the adjusted measures resulting from the econometric approach. The last two columns present the share of the deficit that is attributed to each of the demand constraint and the share of the deficit attributed to other reasons. In line with the discussion in previous section, in urban areas, supply does not seem to constitute a large problem. On the contrary, in rural areas, it is the supply-side gap that dominates. Sixty-five percent of the deficit in coverage in rural areas is attributed to supply-side reasons. In contrast, in urban areas 78% of the deficit in access is attributed to demand-side factors. Nevertheless, even in rural Africa the demand gap is not negligible as it accounts for 35% of the deficit in access.

Next, we turn to the analysis of the constraints representing different aspects of the demand side barriers. Each of the constraints represents the share of the deficit that is attributed to this particular constraint and not the demand gap as a whole as in Wodon et al. (2009).[23] We start by estimating the uptake in areas under the grid using Eq. 2 noting that the table with summary statistics of all variable's used in this model can be found in the appendix (Table 8). Table 1 presents the estimated coefficients from the probit model for all SSA as well as for rural and urban areas respectively. Given that some regressors may be endogenous, these results should be interpreted as correlations. As expected, rural households face

[23] Therefore some of the measures are missing from the bottom panel of Table 2.

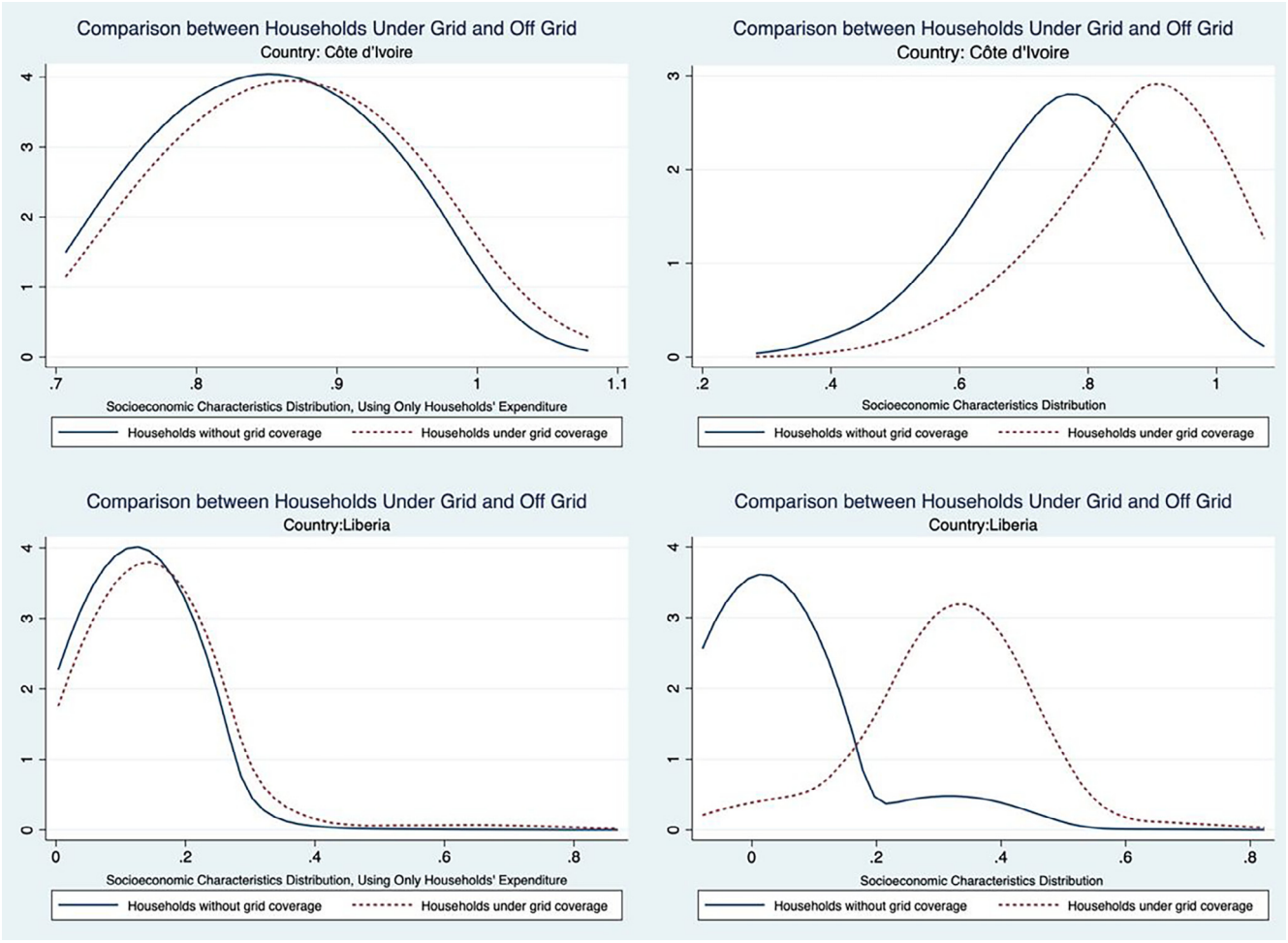


Fig. 12. Comparison between households under and off grid. (*Data source*: Adapted from LSMS, 2010–2016.).

lower connection probabilities. Higher share of connected households in the PSU increases the probability of connection. The share of professionals positively affects the probability of connection in the whole sample and in the urban areas. We also find that the share of individuals in agriculture decreases the probability of connection significantly. At a household level, higher wealth obviously increases the chance of connection. We also confirm the importance of the quality of the dwelling, as measured by the roof quality as well as the importance of regular cash paying income in the rural areas. Next, to quantify the role of each constraint, we perturb each of the measures separately and using the estimated coefficients of Eq. 2, we simulate uptake[24] in areas under the grid. Then, we compute the APDSG as defined in Section 4.1. Lastly, the share of access deficit that is driven by each constraint is computed as $\frac{(APDSG|\text{demandsideconstraint})}{1-A}$. The results are summarized in the bottom panel of Table 2. Note that there are several factors contributing to the demand side in our analysis. Each time we simulate the demand, we perturb only one factor and leave all other factors at their observed value. Therefore the *demand gap* refers to the share of demand that is driven by the particular factor that is altered in a simulation. The estimated demand-side gap due to each of the constraint is of similar magnitude for all three constraint. However, as demonstrated in the last two columns of Table 2 even small changes in the size of the demand gap result in large changes in the share in the access deficit. In urban areas, 25% of the deficit is due to broadly understood affordability constraint. Thirty-eight percent of the deficit can be attributed to low housing quality or lack of permanent cash paying jobs. In rural areas, 21% if the deficit in access can be attributed to lack of sufficient income, while 27% is due to lack of stable income or high quality housing. About 98% of households affected by roof quality constraint are also affected by the cash flow constraint. Additionally, about a half of households constrained by affordability, are also either constrained by cash flows or roof quality. This indicates that often very quite specific needs of the households need to be addressed to increase uptake. The fact that the same households respond to the cash flow and roof quality constraints explain the equal shares of the coverage gap induced by both constraints.

The econometric approach of the decomposition gives us a better sense of the order of magnitude of the various constraints than the statistical approach. However, the extent of the precision of the results depends on the richness of the data. For example, the lack of direct measures of income and consumption is limiting. We address this limitation by providing a third set of results, where, instead of predicting the uptake based on the data and the probit

[24] To estimate additional uptake due to the removal of a constraint, using the model estimates, for each unconnected households, we predict whether it is connected to the grid or not and then we compute the share of households in the covered areas that are predicted to be using grid electricity U^*.

Table 5
Determinants of Uptake.

	OLS			Heckman			
Dep. variable	*Uptake rate*			Second stage *Uptake rate*		First stage *Access to the grid*	
	All	Rural	Urban	All	Rural	All	Rural
Rural	−0.047***			−0.026*		−0.606***	
	(-4.871)			(-2.023)		(-6.888)	
% with reliable electricity	0.106***	0.156***	0.058**	0.090***	0.144***		
	(6.846)	(7.284)	(2.678)	(6.041)	(7.994)		
Access to water	0.127***	0.097***	0.170***	0.111***	0.078***	0.134	0.048
	(11.352)	(7.124)	(8.800)	(8.942)	(6.233)	(1.941)	(0.627)
Sewage	0.089***	0.130***	0.074***	0.048***	0.081***	0.941***	0.454*
	(9.867)	(9.187)	(6.540)	(3.507)	(4.144)	(6.151)	(2.136)
Bank	0.028**	0.024	0.033**	0.029*	0.064**	0.228	0.194
	(2.761)	(1.198)	(2.867)	(1.962)	(3.092)	(1.921)	(1.303)
Market	0.025**	0.045***	−0.007	0.027*	0.050***	0.058	0.035
	(2.688)	(3.280)	(-0.612)	(2.317)	(3.940)	(0.865)	(0.468)
School	0.007	0.037*	−0.021	0.021	0.051**	-0.017	−0.056
	(0.633)	(2.398)	(-1.685)	(1.458)	(3.249)	(-0.200)	(-0.578)
% with secondary schooling	0.141***	0.152***	0.101***	0.061**	0.076**	1.140***	1.042***
	(9.542)	(5.893)	(5.817)	(2.750)	(2.832)	(6.262)	(4.836)
% cash paying job	0.067***	0.100***	0.048*	0.044*	0.118***	0.638***	0.619***
	(4.097)	(4.05)	(2.434)	(2.044)	(4.630)	(4.920)	(4.318)
% with high quality roof	0.202***	0.129***	0.208***	0.086***	0.065**	0.369***	0.334**
	(11.043)	(5.940)	(5.240)	(3.855)	(2.869)	(3.565)	(3.009)
Av. wealth index	0.138***	0.180**	0.091*	0.057	0.095	0.898*	0.904*
	(3.629)	(2.894)	(2.134)	(1.101)	(1.577)	(2.537)	(2.330)
% of president's ethnicity	−0.191***	−0.333***	−0.136***	−0.050	−0.318***	−1.539***	−1.577***
	(-10.512)	(-11.270)	(-6.102)	(-1891)	(-7.102)	(11.035)	(-9.994)
Constant	0.352***	0.348***	0.481***	0.716***	0.534***	0.986***	0.222
	(10.266)	(6.171)	(8.711)	(14.343)	(9.129)	(3.012)	(0.644)
Main energy source hydro						0.508***	0.461***
						(6.129)	(4.495)
Contact with govt. agency						0.054	0.110
						(1.547)	(1.832)
Mills lambda				−0.268***	−0.063*		
				(-12.828)	(-2.002)		
N	3882	1807	2075	4328	2621	4328	2621

Coefficients and *z* statistics in parentheses. * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.
Notes: In all models we also control for other variables across different levels. At country level: GDPPPP, land elevation, population density, Mo Ibrahim Rural sector score, ELF score, governance; at community level: presence of clinic, access to cellular network, paved roads, public transport and the types of economic activity in the community (ratio of agricultural workers, trade workers, upper professionals, security workers, supervisor, clerical workers, skilled workers, unskilled workers) and unemployment rate. The last two column present the results of the first stage of the Heckman procedure. The χ^2 corresponding to the Wald test of joint significance of the exclusion restrictions is 42 in "all" and 29 in "rural" sample indicating that they are important predictors of access. With respect to the validity of the instruments we first include the exclusions in the second stage of the Heckman model to verify that they are insignificant in the main equation. We also estimate the model using maximum likelihood and run an LR test of our specification against a specification with the instruments in the main equation and we fail to reject the null hypothesis that the fit of the model is the same implying that inclusion of the two exclusion restrictions in the main equation does not improve the fit of the model significantly. Even though the variable "contact with government agency" appears not significant, its exclusion threatens the validity of the exclusions.

model reported in Table 1, we instead assume different plausible scenarios.

Given that in SSA, the off-grid areas are more remote and less developed, we first assume that if the grid was expanded there today, the uptake would be two-thirds of the current uptake in the areas already covered. We then present three scenarios: Scenario 1, (respectively 2, and 3) assumes that in the covered areas, 5% of the access deficit (respectively 10%, 15%) can still be attributed to supply-side constraints despite the fact that those households are under the grid. These assumptions reflect the extent of the density of the grid in the community as well as the quality (reliability) of the supply which may also deter uptake. Therefore, the different scenarios reflect varying degrees of the quality of the supply in communities under the grid.

Table 3 summarizes the results by splitting the countries into sub groups defined based on *per capita* income, share of urban population, geographic location and share of renewable energy. Across the three scenarios, even in the most conservative one (Scenario 3), on average, demand-related factors could account for 37% of the access deficit in SSA. In urban areas, demand-side constraints are more salient than supply-side (66% related to 21%) while in rural areas, these two are comparable.

This analysis also underscores the divergence across the countries in the region in the factors that help explaining the electricity gap. Demand considerations are overall more prominent in lower-middle-income countries, compared with low-income countries. The finding suggests that infrastructure build up is initially critical, but becomes less so if economic opportunities are not available to mitigate the affordability issues.

Before proceeding, note that so far we have assumed that a household has access regardless of the quality of the electricity supplied. We now redo the decomposition assuming that for a household to be labeled as connected, this household has to report having electricity at least 50% of the time. The details of this analysis are presented in the appendix, however, it becomes apparent that while demand side barriers are still sizable, the role of the supply side barriers increases reflecting the fact that reliability remains a large and persistent problem across SSA countries. In addition, reliability of electricity services can also affect household's decision to connect to the grid as households in areas with

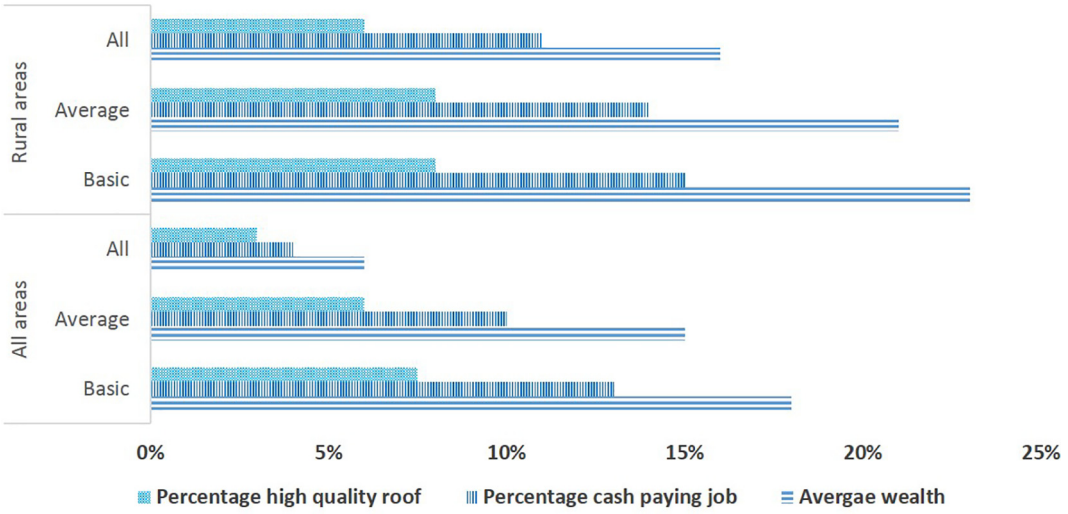


Fig. 13. Marginal effects of selected variables by community infrastructure development level.

more reliable electricity are more likely to connect to the grid as indicated by a strong and positive correlation between household connection rates and reliability of electricity service presented in Fig. 11.

5.2. Uptake in off-grid areas

So far we have focused on households' propensity to connect to the grid in serviced communities. Next, we focus on off grid areas. Table 4 presents the comparison between covered and uncovered areas and confirms that there are significant differences along observable characteristics. For the sake of brevity we only include the characteristics along which substantial differences are observed. Less individuals in covered areas are employed in agriculture or in trading while more hold professional jobs. The covered communities are more likely to have access to piped water, sewage infrastructure, paved roads and public transport. Banks, markets, post offices and clinics are more likely to be present in communities under the grid. The share of individuals with secondary schooling or cash paying jobs is higher in the localities covered by the grid and high quality roofs are more likely in such localities. These crude summary statistics suggest that areas enjoying higher level of infrastructure and economic development are also more likely to have access to the grid. It is also apparent that the share of individuals who think that electricity is an important issue for the government and that the government is handling the issue of electricity well is higher in the communities that are covered by the grid. The dissimilarities also exist at the household level, as evident in Fig. 12. If we consider only household income (household food and non-food expenditure as the proxy), households in serviced and non serviced areas exhibit large similarities, as shown in the left panels of Fig. 12. However, when we add housing quality, dwelling ownership and assets into comparison, the difference between the two groups of households increases significantly as depicted in the right panels of Fig. 12. Such differences at both community and household level confirm our conjecture and further motivate the use of the Heckman two step procedure to factor them in when projecting uptake in currently uncovered areas. We present summary statistics of all variables in the whole sample as well as in the rural sample in the appendix (Table 9). The results of the first stage, summarized in the last two columns of Table 5[25] are of importance to policy makers as they help determine what type of communities are more likely to be connected to the grid. Since the list of covariates is long, for the sake of brevity, we select the variables with significance. Consistently with previous evidence, rural areas are at disadvantage in terms of grid coverage. Overall, it appears that grid coverage comes together with other development efforts and communities, which already enjoy and benefit from better infrastructure are more likely to be covered by the grid.

To verify the relevance of the exclusion restrictions, we conduct a Wald test for joint significance of the two variables in the first step of the estimation. At the bottom of Table 5 we present the results of this test noting that in both cases there is evidence that both variables are significant predictors of access.[26] We proceed with the analysis of marginal effects to better understand the context in which particular characteristics correlate with the grid connection of the locality. We explore different scenarios of infrastructure development at a community level to establish what level of infrastructure development allows communities to benefit the most from other factors in terms of grid connection. We divide communities into three categories. The first category (*Basic*) comprises localities with only basic infrastructure which we consider to be a school and a clinic. The second group (*Average*) contains communities with basic infrastructure and a market, a post office, access to public transport, cell phone reception and access to water and paved roads. The third group (*All*) comprises of communities endowed with all of the above, including a bank and sewage system. We consider changes in three community level characteristics that capture the economic livelihood of the locality: share of households with high quality roof, share of households with cash paying jobs and average wealth of households. Fig. 13 summarizes the results showing that the least developed communities benefit the most from higher fraction of individuals with cash paying jobs, higher fraction of individuals with high quality roof and increase in average wealth. If governments tend to bundle development programs then extra efforts are required in the less developed communities to

[25] We estimate the model using the two step procedure noting that results from maximum likelihood are quantitatively the same.

[26] We have also run a linear probability model and the value of the F test (for joint significance of the exclusion restrictions) is 34 in the "All" sample and 19 in the "Rural" sample which is also suggestive of relevance of exclusions.

obtain grid coverage and that is manifested by the larger marginal effects for all three factors. Next consider the results of the main equation summarized in Table 5. The first three columns present the results of OLS estimation for "All", "Urban" and "Rural" samples separately. The following two columns summarize the results of the second step of the Heckman selection model for the "All" and "Rural" samples. The comparison between OLS and Heckman results reveals substantial selection bias for some covariates manifested in large differences between the magnitudes of the estimated effects. Therefore, we focus on the results of the Heckman model. All infrastructure indicators, as well as the variables capturing households well-being, positively affect community level uptake. This delivers further evidence that economic well-being of a community is crucial for a community to enjoy a high uptake. Hence, it must either precede the presence of grid coverage or, otherwise, electricity must enable individuals to raise their economic well-being.

Among country level indicators, notice that the uptake decreases with the share of population that is of president's ethnicity. Assuming that this variable is a good proxy for political competition, such negative correlation is consistent with the politician efforts along development goals deemed necessary to win elections. The larger the political competition, the more resources may be devoted to maintenance of the infrastructure and, thus, more reliable electricity is available which can very well increase the uptake rates. We do not have sufficient information in the data to confirm such conjecture. However, our results do confirm that

Table 6
Variables in quantitative analysis.

Variable	Definition/explanation/notes	Source
At a community level (PSU):		
Rural area	Indicator variable (1 = rural; 0 = urban)	Afrobarometer
Connection rate	Share of connected households in the PSU	Afrobarometer
Share of households with reliable electricity	Reliable electricity is defined as having electricity at least most of the time	Afrobarometer
Share of households with high quality roof	High quality roof is defined as roof with following materials: Metal, concrete, tin, zinc, tiles, shingle or asbestos	Afrobarometer
Share of respondents:		
thinking electricity is handled well		Afrobarometer
thinking electricity is an important issue for the government		Afrobarometer
employed in agriculture		Afrobarometer
employed in trading occupations		Afrobarometer
employed unskilled occupations		Afrobarometer
employed in skilled occupations		Afrobarometer
employed in clerical occupations		Afrobarometer
employed in supervisory occupations		Afrobarometer
Employed in security		Afrobarometer
employed in upper professional occupations		Afrobarometer
with secondary schooling		Afrobarometer
with cash paying job		Afrobarometer
unemployed		Afrobarometer
Average wealth	Mean wealth index in the community	Afrobarometer
Water	Indicator (1 = community has access to clean water)	Afrobarometer
Roads	Indicator (1 = community has access to paved roads)	Afrobarometer
Sewage	Indicator (1 = community has access to sewage system)	Afrobarometer
Cell	Indicator (1 = community has access to cellular reception)	Afrobarometer
Bank	Indicator (1 = a bank is in the community)	Afrobarometer
Market	Indicator (1 = a market is in the community)	Afrobarometer
Transport	Indicator (1 = community has access to public transport)	Afrobarometer
Post office	Indicator (1 = a post office is in the community)	Afrobarometer
Clinic	Indicator (1 = a clinic is in the community)	Afrobarometer
School	Indicator (1 = a school is in the community)	Afrobarometer

Table 7
Variables in quantitative analysis.

Variable	Definition/explanation/notes	Source
At a country level:		
GDP per capita		World Development Indicators
Polity IV		Polity IV Index on Democracy
Share of individuals of president's ethnicity		Afrobarometer
Ethno linguistic fragmentation		Alesina et al. (2003)
Mo Ibrahim infrastructure index		Rotberg and Gisselquist (2008)
Mo Ibrahim rural index		Rotberg and Gisselquist (2008)
Population density		World Development Indicators
At a household or respondent level:		
Wealth index quintile*	Categorical variable indicating the household position in the wealth distr.	Afrobarometer
Secondary schooling	Indicator whether respondent completed secondary schooling or not	Afrobarometer
Roof	Indicator whether the households has high quality roof	Afrobarometer
Age	Age of the respondent	Afrobarometer
Female	Gender of the respondent; indicator variable (1 = female; 0 = men)	Afrobarometer
Urban	Locality of the households: indicator variable (1 = urban; 0 = rural)	Afrobarometer
Household size	Number of adult individuals residing in the dwelling	Afrobarometer

Notes: *Variable *Wealth index* is constructed from the Afrobarometer data based on the following variables at the household level: dummy variables indicating possession of a phone,a radio, a TV, a motor, a vehicle or a motorcycle; dummy variables indicating the presence of a toilet and access to running water, self-reported economic situation of the household (absolute and relative to neighbours) and dummy variable for receiving remittances and a dummy variable indicating whether the household often experiences lack of cash. All components have equal weight and the score is transformed to be on a on 0–1 scale.

Table 8
Summary statistics of variables used in decomposition analysis.

	All		Rural		Urban	
Connected to the grid	0.74	(0.44)	0.60	(0.49)	0.87	(0.33)
Share connected in the PSU	0.74	(0.31)	0.59	(0.33)	0.87	(0.22)
Rural	0.48	(0.50)				
Age	35.89	(13.45)	36.66	(13.96)	35.19	(13.32)
Female	0.50	(0.50)	0.51	(0.50)	0.50	(0.50)
Household size	3.94	(2.40)	4.10	(2.49)	3.79	(2.29)
High quality roof	0.84	(0.37)	0.73	(0.44)	0.94	(0.24)
Secondary schooling or above	0.37	(0.48)	0.28	(0.45)	0.46	(0.50)
Cash paying job	0.29	(0.45)	0.25	(0.43)	0.33	(0.46)
Share employed in agriculture	0.15	(0.36)	0.26	(0.44)	0.06	(0.24)
Share employed in upper professional occupations	0.22	(0.42)	0.15	(0.36)	0.29	(0.45)

Notes: Standard deviations in brackets.

Table 9
Summary statistics of variables used in the Heckman selection model.

	Serviced households				All households			
	All		Rural		All		Rural	
Rural	0.46	(0.50)			0.59	(0.49)		
Share of households with:								
access to clean water	0.68	(0.47)	0.45	(0.50)	0.58	(0.50)	0.38	(0.50)
access to paved roads	0.59	(0.49)	0.41	(0.49)	0.51	(0.50)	0.35	(0.48)
access to sewage system	0.39	(0.49)	0.11	(0.31)	0.29	(0.45)	0.07	(0.25)
cellular reception	0.92	(0.26)	0.86	(0.35)	0.91	(0.29)	0.86	(0.35)
access to a bank	0.31	(0.46)	0.13	(0.34)	0.24	(0.43)	0.10	(0.30)
access to a market	0.69	(0.46)	0.60	(0.49)	0.65	(0.46)	0.58	(0.50)
access to public transport	0.88	(0.33)	0.80	(0.40)	0.81	(0.39)	0.73	(0.45)
access to a post office	0.26	(0.44)	0.13	(0.33)	0.20	(0.40)	0.10	(0.29)
access to a clinic	0.60	(0.49)	0.52	(0.50)	0.56	(0.50)	0.50	(.50)
access to a school	0.86	(0.35)	0.84	(0.37)	0.86	(0.34)	0.85	(0.35)
with high quality roof	0.85	(0.26)	0.74	(0.32)	0.77	(0.32)	0.64	(0.35)
with reliable electricity	0.55	(0.38)	0.43	(0.40)	0.55	(0.38)	0.43	(0.40)
Share of respondents:								
that thinks that government handles electricity well	0.62	(0.25)	0.61	(0.26)	0.59	(0.26)	0.56	(0.26)
that thinks electricity is important for government	0.17	(0.2)	0.17	(0.21)	0.16	(0.21)	0.15	(0.20)
with at least secondary schooling	0.38	(0.29)	0.27	(0.25)	0.30	(0.29)	0.20	(0.23)
with cash paying job	0.42	(0.27)	0.38	(0.27)	0.38	(0.28)	0.33	(0.28)
unemployed	0.31	(0.22)	0.30	(0.23)	0.29	(0.24)	0.27	(0.25)
of president's ethnicity	0.25	(0.26)	0.25	(0.27)	0.29	(0.29)	0.32	(0.30)
employed in agriculture	0.06	(0.24)	0.07	(0.26)	0.09	(0.29)	0.11	(0.32)
employed in trading	0.05	(0.23)	0.06	(0.24)	0.06	(0.24)	0.08	(0.27)
employed in unskilled occupations	0.09	(0.28)	0.09	(0.29)	0.10	(0.30)	0.11	(0.31)
employed in skilled occupations	0.12	(0.32)	0.12	(0.32)	0.12	(0.33)	0.12	(0.32)
employed in clerical occupations	0.03	(0.18)	0.03	(0.18)	0.03	(0.16)	0.03	(0.16)
employed in supervisory occupations	0.03	(0.17)	0.03	(0.16)	0.03	(0.16)	0.02	(0.15)
employed in security	0.05	(0.21)	0.04	(0.20)	0.04	(0.21)	0.04	(0.19)
employed in upper professional occupations	0.09	(0.29)	0.09	(0.29)	0.08	(0.27)	0.07	(0.26)
employed in other occupations	0.21	(0.41)	0.19	(0.39)	0.20	(0.40)	0.18	(0.38)
Average wealth	0.20	(0.09)	0.19	(0.09)	0.19	(0.09)	0.19	(0.09)
GDP per capita	4956	(4593)	4318	(3942)	4104	(4253)	3197	(3367)
Polity IV index	4.74	(4.43)	4.21	(4.82)	4.55	(4.14)	4.15	(4.19)
Ethno-linguistic fragmentation	0.72	(0.22)	0.71	(0.24)	0.69	(0.24)	0.66	(0.26)
Mo Ibrahim infrastructure score	41.47	(17.72)	39.30	(16.73)	39.23	(16.38)	36.61	(14.34)
Mo Ibrahim rural score	52.52	(11.98)	52.96	(10.67)	51.76	(12.68)	51.41	(12.54)
Population density	80.69	(62.13)	86.91	(64.65)	88.52	(77.68)	96.99	(86.54)

Notes: Standard deviations in brackets.

uptake increases with a higher share of households enjoying electricity at least most of the time. Also, the evidence from our field work suggests that frequent blackouts experienced in the community can deter unconnected households from connecting.

For electrification efforts to be successful, the uptake of electricity in targeted communities needs to be high. However, our results show that the average truncation effect in the sample of communities with access to the grid in the rural areas is -0.74 (*mean inverse mills ratio* $\times(0.063)$) indicating that the uptake rate in an average rural community is 4.6% ($[exp(av.truncation) - 1] \times 100$) lower than it would be in a randomly selected rural community. This suggests that the selection of communities for grid expansions does not fully utilize the underlying potential and more effective targeting strategies need to be developed to ensure higher uptake rates.

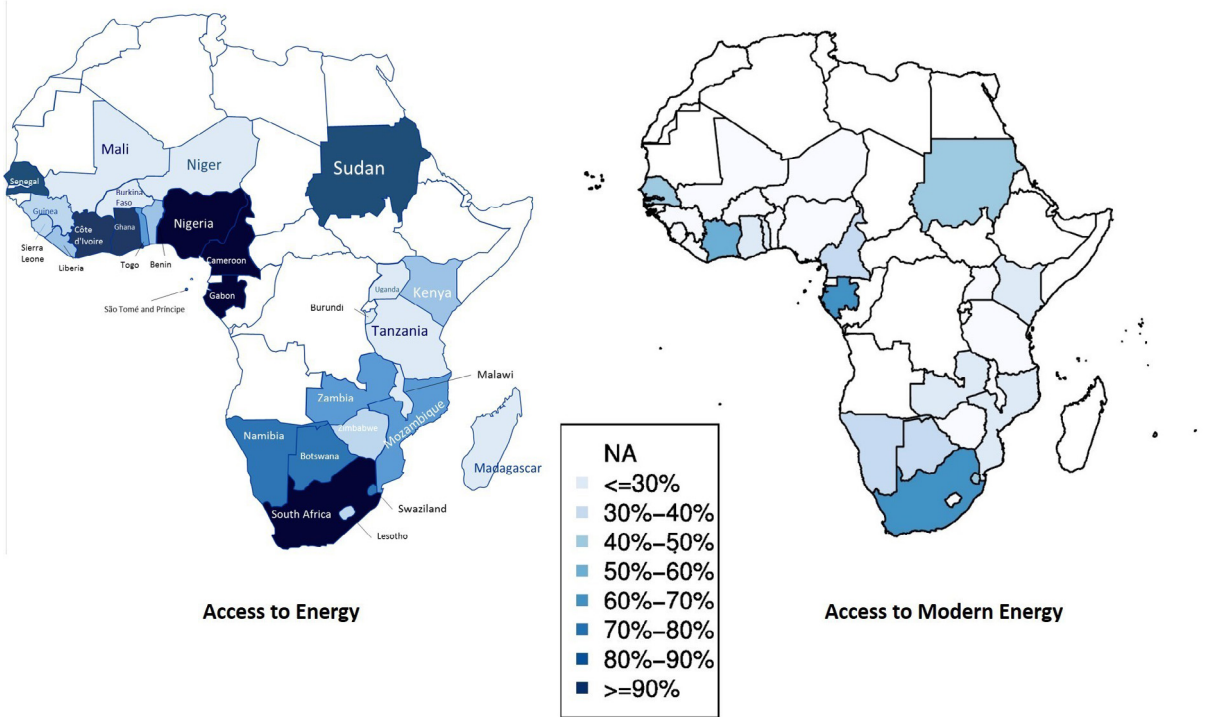


Fig. 14. Access to modern electricity. (*Data source*: Afrobarometer, 2014/15.).

Table 10
Access to modern electricity: demand–supply decomposition for Sub-Saharan Africa

	PDSG	SSG	PSSG	MDSG	% deficit "demand"	% deficit rest
All Africa	0.09	0.43	0.35	0.08	0.17	0.83
Urban Africa	0.10	0.32	0.27	0.05	0.24	0.76
Rural	0.12	0.47	0.37	0.10	0.20	0.80

Notes: PDSG denotes the pure demand side gap, SSG denotes the supply side gap, PSSG denotes pure supply side gap and MDSG denotes the mixed demand side gap. See appendix B for details of computation. Access to modern electricity is defined as having electricity at least 50% of the time.

6. Conclusions

The large infrastructure gap in Africa has led to disproportionate focus on supply issues in the electrification efforts. We present extensive evidence on the demand-side constraints that explain the persistently low household electricity access in a large number of African countries. We show that demand constraints may be as important as the supply side constraints. We identified household income levels, income predictability, housing quality, and the reliability of electricity as key factors affecting uptake when the service is available. Furthermore, household perceive electricity as a vehicle for more productive income generating activities, which often require complementary inputs such as access to capital to start income generating activities. Several policy implications stem from this research:

i. Electricity sector planing should incorporate demand and uptake at the planning stage to ensure that the gradual roll-out of electrification generate enough uptake to sustain the financial viability of the utilities and service providers.
ii. Several technical solutions like pre-paid metering (to address income irregularity), ready boards (to address housing quality), or reducing the upfront connection charges can help with additional household uptake on the margins.
iii. Electrification policies should be centered around income generation and economic activities as is expected by the household. This focus requires providing complementary investments alongside electricity in order to speed-up the economic impact on households. Utilities too could innovate, for example, in providing services beyond electricity alone (e.g. financing facilities) and boost access to productive electrical appliances.

Given the resource constraint to justify a blanket electrification of many African countries, future research on this issue on the continent should focus on ways to ensure that electricity supports growth and income generation in order to support a financially stable sector. Therefore, key research questions should focus on understanding complements and conditions under which electricity access contribute to raising household income and livelihood.

CRediT authorship contribution statement

Moussa P. Blimpo: Conceptualization, Methodology, Investigation, Writing - review & editing, Supervision. **Agnieszka Postepska:** Conceptualization, Validation, Investigation, Formal analysis, Writing - original draft, Writing - review & editing, Software. **Yanbin Xu:** Visualization, Investigation, Data curation, Software, Writing - review & editing.

Acknowledgement

We are grateful to Ruifan Shi for excellent research assistance. Thanks to participants at the authors' workshop in May 2016 at the World Bank. Thanks to participants in conferences at Oxford University (CSAE annual conference, March 2018) and Duke University (SETI, May 2018), for their comments and suggestions. We also thank Malcolm Cosgrove-Davies, Wilfried A. Kouame,Subodh Mathur, Kabir Malik, Justice Tei Mensah, and Aly Sanoh for their comments. We are especially grateful for the comments of the two anonymous referees. The authors assume all responsibility for any errors. This paper was a part of a series of background papers to the "Regional Study on Electricity Access in Sub-Saharan Africa" produced by the Office of the Chief Economist in Africa Region (AFRCE), World Bank, under the project code P156903. The findings, interpretations, and conclusions expressed in this paper are entirely those of the authors. They do not necessarily represent the views of the International Bank for Reconstruction and Development/World Bank and its affiliated organizations, or those of the Executive Directors of the World Bank or the governments they represent

Appendix A. Data sources and variables definitions

The survey based data sets are complemented with country level indicators such as Demographic and Health Surveys (DHS), World Development Indicators (WDI), Mo Ibrahim Index on Africa Governance (Rotberg & Gisselquist, 2008), Polity IV Index on Democracy and Ethno-Linguistic Fragmentation Score (Alesina, Devleeschauwer, Easterly, Kurlat, & Wacziarg, 2003). Electrification rates from DHS and WDI data are used as a robustness check. Mo Ibrahim Index on Governance consists of four categories: Safety and Rule of Law, Participation and Human Rights, Sustainable Economic Opportunity and Human Development. These categories comprise 14 sub-categories, consisting of 102 indicators, from 35 data sources. This analysis uses Infrastructure Index and Rural Sector Index (two sub-categories of Sustainable Economic Opportunity). The former consists of six indicators including transport infrastructure, digital and IT infrastructure, access to drinking water, reliability of electricity supply, satisfaction with provision of water and sanitation services and enabling environment for infrastructure development. The later is composed by access to rural land and water for agriculture, rural business climate, government investment in rural development, access to agricultural research and extension services, budgetary balance of agriculture policy, participation of rural organizations, promotion of gender parity in rural organizations and rural accountability and transparency. These indexes range from 0 to 100, noting that higher score implies better performance.

Polity IV Index on Democracy captures regime authority spectrum on a 21-point scale ranging from −10 (hereditary monarchy) to +10 (consolidated democracy). Ethno-Linguistic Fragmentation Score measures the degree of ethnic, linguistic and religious heterogeneity in various countries, enabling researchers to examine the effects of fictionalization on different outcomes. Moreover, to examine what factors affect the presence of grid in community, we utilize georeferenced data on electricity transmission network in Africa from Africa Infrastructure Country Diagnostic database (AICD), a project by World Bank designed to understand physical infrastructure in Africa.

Country level data from the last round of the survey from year 2010 to 2017 of the Living Standards Measurement Study (LSMS) are adopted to analyze the characteristics of the households in covered and uncovered areas. LSMS is a household survey program conducted by World Bank's Development Data Group to measure varied dimension of living situation of the households.

The following tables contain further details on all variables employed in the quantitative analysis. Table 6 and Table 7 contain variables definitions, sources and some additional explanation. Table 8 contains summary statistics of all variables used in the probit model in the decomposition (Eq. 2) while Table 9 contains the summary statistics of all variables used in the Heckman selcection model.

Appendix B. Details of the decomposition

This section aims to append the empirical methods section with the detailed description of both the statistical approach and the econometric approach to the demand–supply decomposition, together with a numerical example (for details see Wodon et al. (2009) but note the difference in definitions). Consider the following hypothetical example. Assume the following: $A = 0.15, C = 0.05, U = 0.3$ (noting that $C = A \times U$). Recalling that A denotes access, U denotes uptake and C denotes coverage, all objects of interest in the statistical approach are obtained using the following equations:

$$\begin{aligned}
&\textit{Pure demand side gap} && : \quad PDSG = C \times (1 - U) = 0.04 \\
&\textit{Supply side gap} && : \quad SSG = 1 - C = 0.95 \\
&\textit{Pure supply side gap} && : \quad PSSG = (1 - C) \times U = 0.29 \\
&\textit{Mixeddemand} - \textit{sidegap} && : \quad MDSSG = SSG \times (1 - U) = 0.67
\end{aligned}$$

The share of the access deficit that is due to demand is $\frac{PDSG}{1-A} = 0.05$, the share of the access deficit that is due to supply side constraint equals $\frac{PSSG}{1-A} = 0.34$ and the remaining is due to mixed demand–supply factor and is equal to $\frac{MDSSG}{1-A} = 0.79$.

In the econometric approach we first estimate Eq. 2 and then, for each constraint that we consider, we simulate what the uptake among unconnected households (U^*) would be if that constraint is not binding, i.e. we lift everyone to the top quintile of the wealth distribution **or** we assume all households have high quality roof **or** we assume at least one member of the household has a steady cash paying job. More specifically, based on the predicted probabilities we classify unconnected households to be either connected or not and add the households predicted to be connected to the connected households to obtain U^*. Let us assume that $U^* = 0.4$. Subsequently, we perform the decomposition using the following equations:

$$\begin{aligned}
&\textit{Adjusted pure demand side gap} && : \quad APDSG = C \times (U^* - U) = 0.005 \\
&\textit{Adjusted supply side gap} && : \quad ASSG = 1 - A \times U^* = 0.94 \\
&\textit{Adjusted pure supply side gap} && : \quad APSSG = (1 - A \times U^*) \times U^* = 0.38 \\
&\textit{Adjusted mixed demand-side gap} && : \quad AMDSSG = (1 - A \times U^*) \times (1 - U^*) = 0.56
\end{aligned}$$

The corresponding shares in access deficit due to demand, supply or mixed constraints can be obtained in the similar way as in the statistical approach.

Appendix C. Decomposition under different definition of access

This appendix presents the results of the statistical approach of the demand–supply decomposition where we restrict the sample to areas with reliable electricity. Areas are defined as having access to reliable electricity if at least 50% of the household report having electricity at least most of the time (Afrobarometer data). Fig. 14 presents the percentage of households in Sub-Saharan countries which have access to modern energy as defined. Only in South Africa, three in every four households are connected to reliable electricity (the household has electricity most of the time or always). In contrast, in Burundi and Guinea only 4% of households are connected to reliable electricity, huge disparity from 11% and 37% access rate under the definition that does not account for reli-

ability respectively. On average, only 33% of households in Sub-Saharan Africa are located in areas covered by reliable electricity. Rural Africa remains at a disadvantaged position with only 18% of households in sun covered areas. In urban areas the situation is better but still less than 1 in every 2 households is located in areas covered by reliable electricity.

As we restrict the sample to areas with reliable electricity the size of the supply gap increases across the board as illustrated in Table 10. It is especially the case in urban areas where the pure supply-side gap increased from 0.03 to 0.27 and the corresponding deficit in access due to supply from 22% to 76%. These results suggests that accounting for reliability of available electricity is crucial to correctly quantify the demand and supply side deficit. It also underscores the need for appropriate infrastructure improvement to guarantees reliable electricity in electrified areas, encouraging high uptake rates. Furthermore, it calls for a measure of access to electricity that can capture different dimensions of the service provided (capacity, safety and quality), which has been echoed by MTF project[27] (Bhatia & Angelou, 2015), an excellent attempt to change the way we think about energy access and to generate new data to help us better understand this issue across Sub-Saharan Africa.

References

Abdullah, S., & Jeanty, P. W. (2011). Willingness to pay for renewable energy: Evidence from a contingent valuation survey in Kenya. *Renewable and Sustainable Energy Reviews, 15*(6), 2974–2983.

Ahlborga, H., Boräng, S. C., Jagers & Söderholmd, P. (2015). Provision of electricity to african households: The importance of democracy and institutional quality. *Energy Policy, 87*, 125–135.

Alesina, A., Devleeschauwer, A., Easterly, W., Kurlat, S., & Wacziarg, R. (2003). Fractionalization. *Journal of Economic Growth, 8*(2), 155–194.

Banerjee, S., Wodon, Q., Diallo, A., Pushak, T., Uddin, E., Tsimpo, C., & Foster, V. (2008). *Access, affordability, and alternatives: Modern infrastructure services in Africa*. Washington, DC: World Bank.

Bensch, G., Grimm, M., Huppertz, M., Langbein, J., & Peters, J. (2018). Are promotion programs needed to establish off-grid solar energy markets? Evidence from Rural Burkina Faso. *Renewable and Sustainable Energy Reviews, 90*, 1060–1068.

Bhatia, M., & Angelou, N. (2015). *Beyond connections energy access redefined*. Washington, DC: World Bank.

Blimpo, M., McRae, S., & Steinbuks, J. (2017). Electricity Access Charges and Tariff Structure in Sub-Saharan Africa. Working Paper..

Blimpo, M., Mensah, J., Opalo, K., & Shi, R. (2018). Electricity Provision and Tax Mobilization in Africa. World Bank Policy Research Working Paper 8408..

CGD (2017). *More than a light bulb: Five recommendations to make modern energy access meaningful for people and prosperity*. Center for Global Development.

Chaplin, D., Mamun, A., Protik, A., Schurrer, J., Vohra, D., Bos, K., & Cook, T. (2017). Grid electricity expansion in Tanzania by MCC: Findings from a rigorous impact evaluation. *Mathematica Policy Research*.

Coville, A., Orozco-Olvera, V., & Reichert, A. (2019). Paying Attention to Profitable Investments: Experimental Evidence from Renewable Energy Markets. Masami..

Dasso, R., Fernandez, F., & Nopo, H. (2015). *Electrification and educational outcomes in Rural Peru*. Institute of Labor Economics.

Dinkelman, T. (2011). The effects of rural electrification on employment: New evidence from South Africa. *American Economic Review, 101*(7), 3078–3108.

Eberhard, A., Rosnes, O., Shkaratan, M., & Vennemo, H. (2011). *Africa's power infrastructure: Investment, integration, efficiency*. Washington, DC: World Bank.

Foster, V., & Araujo, C. (2004). Does Infrastructure Reform Work for The Poor? A Case Study from Guatemala. World Bank Policy Research Working Paper 3185..

Grimm, M., Munyehirwe, A., Peters, J., & Sievert, M. (2016). *A first step up the energy ladder? Low cost solar kits and household's welfare in rural Rwanda*. Washington, DC: World Bank.

Haanyika, C. M. (2006). Rural electrification policy and institutional linkages. *Energy Policy, 34*(17), 2977–2993.

Heckman, J. (1976). The Common Structure of Statistical Models of Truncation, Sample Selection and Limited Dependent Variables and a Simple Estimator for Such Models. in Annals of Economic and Social Measurement..

IEA. (2014). Africa Energy Outlook. Abuja, Nigeria: International Energy Agency. (World Energy Outlook Special Report)..

IEA. (2017). World Energy Outlook..

Jacome, V., Klugman, N., Wolfram, C., Grunfeld, B., Callaway, D., & Ray, I. (2019). Power quality and modern energy for all. *Proceedings of the National Academy of Sciences, 116*(33), 16308–16313.

Kalra, P.K., Shekhar, R., & Shrivastava, V.K. (2007). Electrification and Bio-Energy Options in Rural India. 3IE Network (India) and Infrastructure Development Finance Company (India), eds., India infrastructure report..

Karekezi, S., & Kimani, J. (2002). Status of power sector Seform in Africa: Smpact on the poor. *Energy Policy, 30*(11–12), 923–994.

Khennas, S. (2012). Understanding the political economy and key drivers of energy access in addressing national energy access priorities and policies: African perspective. *Energy Policy, 47*, 21–26.

Kielmann, K., Cataldo, F., & Seeley, J. (2012). *Introduction to qualitative research methodology: A training manual*. United Kingdom: Department for International Development (DfID).

Kowsari, R., & Zerriffi, H. (2011). Three dimensional energy profile: A conceptual framework for assessing household energy use. *Energy Policy, 39*(12), 7505–7517.

Lee, K., Brewer, E., Christiano, C., Meyo, F., Miguel, E., Podolsky, M., & Wolfram, C. (2016). Electrification for "Under Grid" households in Rural Kenya. *Development Engineering, 1*, 26–35.

Lee, K., Miguel, E., & Wolfram, C. (2016). Appliance ownership and aspirations among electric grid and home solar households in Rural Kenya. *American Economic Review, 106*(5), 89–94.

Lee, K., Miguel, E., & Wolfram, C. (2016b). Experimental Evidence on the Demand for and Costs of Rural Electrification (Tech. Rep.). National Bureau of Economic Research Working Paper 22292..

Lenz, L., Munyehirwe, A., Peters, J., & Sievert, M. (2017). Does large-scale infrastructure investment alleviate poverty? Impacts of Rwanda's electricity access roll-out program. *World Development, 89*, 88–110.

Lipscomb, M., Mobarak, A. M., & Barham, T. (2013). Development effects of electrification: Evidence from the topographic placement of hydropower plants in Brazil. *American Economic Journal: Applied Economics, American Economic Association, 5*(2), 200–231.

Marshall, M. N. (1996). Sampling for qualitative research. *Family Practice, 13*, 522–525.

Mensah, J.T. (2018). Jobs! Electricity Shortages and Unemployment in Africa. World Bank Policy Research working paper 8415..

Moner-Girona, M., Bódis, K., Huld, T., Kougias, I., & Szabó, S. (2016). Universal access to electricity in Burkina Faso: Scaling-up renewable energy technologies. *Environmental Research Letters, 11*(8) 084010.

Niez, A. (2010). Comparative Study on Rural Electrification Policies in Emerging Economies. IEA Energy Papers..

Peters, J., & Sievert, M. (2016). Impacts of rural electrification revisited–the African context. *Journal of Development Effectiveness*, 1–19.

Peters, J., Sievert, M., & Toman, M. A. (2019). Rural electrification through mini-grids: Challenges ahead. *Energy Policy, 132*, 27–31.

Rosnes, O., & Shkaratan, M. (2011). *Africa's power infrastructure: Investment, integration, efficiency*. Washington, DC: World Bank.

Rotberg, R., & Gisselquist, R. M. (2008). *Strengthening African Governance*. Cambridge, MA: Harvard Kennedy School.

Trimble, C., Kojima, M., Perez, I.A., & Mohammadzadeh, F. (2016). Financial Viability of Electricity Sectors in Sub-Saharan Africa: Quasi-Fiscal Deficits and Hidden Costs. World Bank Policy Research Working Paper 7788..

van der Kroon, B., Brouwer, R., & van Beukering, P. J. (2013). The energy ladder: Theoretical myth or empirical truth? Results from a meta-analysis. *Renewable and Sustainable Energy Reviews, 20*, 504–513.

Van de Walle, D.P., Ravallion, M., Mendiratta, V., & Koolwal, G.B. (2013). Long-term Impacts of Household Electrification in rural India. World Bank Policy Research Working Paper 6527..

WBG (2012). *World Development Indicators 2012*. Washington, DC: World Bank.

Wodon, Q. T., Banerjee, S. G., Diallo, A. B., & Foster, V. (2009). Is Low Coverage of Modern Infrastructure Services in African Cities Due to Lack of Demand or Lack of Supply? World Bank Policy Research Working Paper Series.

[27] Multi-tier framework